

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 11 — Fiscal Policies in a Growth Model

All 20 end-of-chapter exercises, 11.1–11.20 (pp. 449–470)

What is in this file

The exercises fall into four groups: tax reforms and a capital levy (11.1, 11.2, 11.4, 11.14); planning problems, steady states and small open economies (11.3, 11.5, 11.6, 11.8, 11.9, 11.13, 11.18); reading paths of consumption, capital and interest rates as the outcome of foreseen or unforeseen fiscal policy (11.7, 11.10–11.12, 11.15–11.17); and the “Big K , little k ” logic behind the shooting algorithm (11.19, 11.20).

The companion notebook `ls_ch11.ipynb` (same folder) computes every number quoted here and checks each one by an independent route with an `assert`: Newton on the whole equilibrium path against the shooting algorithm of section 11.6.3 and the linear approximation (11.10.8); the stable root against formula (11.10.16); value-function iteration against closed forms; the household’s own problem at equilibrium prices against the aggregate path. Where the book gives no numbers we use the benchmark of section 11.9, stated below.

Question boxes reproduce the full statement. Blue boxes are odd-numbered exercises, purple boxes even-numbered ones.

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Chapter 11 Fiscal Policies in a Growth Model

Chapter 11 Overview

Economy (11.2). Preferences $\sum_t \beta^t U(c_t, 1 - n_t)$; technology $g_t + c_t + k_{t+1} \leq F(k_t, n_t) + (1 - \delta)k_t$ (11.2.3). Household budget, all trades at time 0 (11.2.4):

$$\sum_t q_t \{(1 + \tau_{ct})c_t + k_{t+1} - (1 - \delta)k_t\} \leq \sum_t q_t \{\eta_t k_t - \tau_{kt}(\eta_t - \delta)k_t + (1 - \tau_{nt})w_t n_t - \tau_{ht}\};$$

government budget $\sum_t q_t g_t \leq \sum_t q_t \{\tau_{ct}c_t + \tau_{kt}(\eta_t - \delta)k_t + \tau_{nt}w_t n_t + \tau_{ht}\}$ (11.2.5).

Equilibrium conditions. No arbitrage in capital (11.5.2): $q_t/q_{t+1} = (1 - \tau_{k,t+1})(\eta_{t+1} - \delta) + 1$. Household FOC (11.5.5a): $\beta^t U_{1t} = \mu q_t(1 + \tau_{ct})$. Firm (11.5.7): $\eta_t = F_{kt}$, $w_t = F_{nt}$. With inelastic labour ($n = 1$, $f(k) = F(k, 1)$) these give the Euler equation (11.6.3)

$$u'(c_t) = \beta u'(c_{t+1})\bar{R}_{t+1}, \quad \bar{R}_{t+1} = \frac{1 + \tau_{ct}}{1 + \tau_{c,t+1}} \left[(1 - \tau_{k,t+1})(f'(k_{t+1}) - \delta) + 1 \right] \quad (11.6.8e),$$

with $c_t = f(k_t) + (1 - \delta)k_t - g_t - k_{t+1}$ (11.6.1), k_0 given and the terminal condition (11.5.3). Steady state (11.6.7): $f'(k) = \delta + \rho/(1 - \tau_k)$, $\beta = 1/(1 + \rho)$, $\bar{R} = 1 + \rho$ (11.6.10). Prices (11.6.8b–d): $q_t = \beta^t u'(c_t)/(1 + \tau_{ct})$, $\eta_t = f'(k_t)$, $w_t = f(k_t) - k_t f'(k_t)$. With CRRA utility (11.6.9): $\log(c_{t+1}/c_t) = \gamma^{-1} \log \beta + \gamma^{-1} \log \bar{R}_{t+1}$.

Term structure (11.3). $m_{t,t+1} = q_{t+1}/q_t = R_{t,t+1}^{-1}$, $r_{t,t+1} = -\log m_{t,t+1}$; $q_t = q_0 \exp(-tr_{0,t})$ with $r_{0,t} = t^{-1}(r_{0,1} + \dots + r_{t-1,t})$ (11.3.2). Government debt (11.4.3): $B_t = \sum_{s>t} (q_s/q_t)(T_s - g_s)$, and $g_t + R_{t-1,t}B_{t-1} = T_t + B_t$ (11.4.2).

What distorts (11.8). Lump-sum taxes: only their present value matters (Ricardian equivalence). With inelastic labour, constant τ_c and τ_n do not distort; changes in τ_c over time and τ_k do.

Shooting (11.6.3): guess c_0 , iterate (11.6.1) and (11.6.3) forward, adjust c_0 until $k_S \approx \bar{k}$.

Linear approximation (11.10): $\phi_0 k_{t+2} + \phi_1 k_{t+1} + \phi_2 k_t = A_0 + A_1 z_t + A_2 z_{t+1}$, $\phi(L) = \phi_0(1 - \lambda_1 L)(1 - \lambda_2 L)$, $|\lambda_2| < 1 < |\lambda_1|$, and (11.10.8)

$$k_{t+1} = \lambda_2 k_t - \lambda_2 \phi_2^{-1} \sum_{j \geq 0} \lambda_1^{-j} [A_0 + A_1 z_{t+j} + A_2 z_{t+j+1}] :$$

feedback coefficient λ_2 , feedforward discount λ_1^{-1} ; when the equilibrium solves a planning problem, $\lambda_1 \lambda_2 = 1/\beta$, so $\lambda_1^{-1} = \beta \lambda_2$ (11.10.1).

Elastic labour (11.12), $U = \ln c + B(1 - n)$: $c_{t+1} = \beta \bar{R}_{t+1} c_t$, $B c_t = \frac{1 - \tau_{nt}}{1 + \tau_{ct}} F_n(k_t, n_t)$, $k_{t+1} = F(k_t, n_t) + (1 - \delta)k_t - g_t - c_t$ (11.12.15).

Misprint in the chapter: formula (11.10.16)

As printed, $\lambda_2 = \gamma / (a_1 \gamma^{-1} + a_2 + a_3(\gamma^{-1} + a_4 \gamma^{-2} + a_5)^{1/2})$ gives $\lambda_2 = 3.53 > 1$ at $\gamma = 2$ and grows without bound in γ . The formula that matches the stated constants and figure

11.10.1 is

$$\lambda_2 = \frac{\gamma}{a_1\gamma + a_2 + (a_3\gamma + a_4\gamma^2 + a_5)^{1/2}},$$

with $a_1 = \frac{1+\beta}{2}$, $a_2 = \frac{\beta A}{2}$, $a_3 = \frac{\beta(1+\beta)A}{2}$, $a_4 = \left(\frac{1-\beta}{2}\right)^2$, $a_5 = a_2^2$ and $A = \beta\bar{c}|f''(\bar{k})|$. It is derived in exercise 11.7(c) and checked there against the root of the characteristic polynomial.

Conventions and parameters used throughout

Benchmark (section 11.9) wherever the book gives no numbers: $f(k) = k^\alpha$ ($z = A = 1$), $\alpha = .33$, $\delta = .2$, $\beta = .95$ ($\rho = .0526$), $u(c) = c^{1-\gamma}/(1-\gamma)$ with $\gamma = 2$, $g = .2$. Then $\bar{k} = 1.489956$ and $\bar{c} = f(\bar{k}) - \delta\bar{k} - g = .642645$. Deviations are stated where they are used (and why). Prices are normalised by $q_0 = 1$; $r_{t,t+1} = \log(q_t/q_{t+1})$. “Newton” means solving the Euler equations for $k_1, \dots, k_{150}$ jointly with $k_{151} = \bar{k}$; “shooting” is the algorithm of section 11.6.3 with bisection on c_0 . Welfare is $\sum_t \beta^t u(c_t)$; a consumption-equivalent gain λ solves $W((1+\lambda)c) = W'$.

Tax Reforms, a Productivity Shift and a Capital Levy (11.1–11.4)

Exercise 11.1 Tax reform: I

Question

p. 449. Consider the following economy populated by a government and a representative household. There is no uncertainty, and the economy and the representative household and government within it live forever. The government consumes a constant amount $g_t = g > 0$, $t \geq 0$. The government also sets sequences for two types of taxes, $\{\tau_{ct}, \tau_{ht}\}_{t=0}^\infty$. Here τ_{ct}, τ_{ht} are, respectively, a possibly time-varying flat-rate tax on consumption and a time-varying lump-sum or “head” tax. The preferences of the household are ordered by $\sum_{t=0}^\infty \beta^t u(c_t)$, where $\beta \in (0, 1)$ and $u(\cdot)$ is strictly concave, increasing, and twice continuously differentiable. The representative household is endowed with one unit of labor each period and does not value leisure. The feasibility condition in the economy is

$$g_t + c_t + k_{t+1} \leq f(k_t) + (1 - \delta)k_t$$

where k_t is the stock of capital owned by the household at the beginning of time t and $\delta \in (0, 1)$ is a depreciation rate. At time 0, there are complete markets for dated commodities. The household faces the budget constraint:

$$\sum_{t=0}^{\infty} \{q_t[(1 + \tau_{ct})c_t + k_{t+1} - (1 - \delta)k_t]\} \leq \sum_{t=0}^{\infty} q_t\{\eta_t k_t + w_t - \tau_{ht}\}$$

where we assume that the household inelastically supplies one unit of labor, and q_t is the price of date t consumption goods measured in the numeraire at time 0, η_t is the rental rate of date t capital measured in consumption goods at time t , and w_t is the wage rate of date t labor measured in consumption goods at time t . Capital is neither taxed nor subsidized.

A production firm rents labor and capital. The production function is $f(k)n$, where $f' > 0$, $f'' < 0$. The value of the firm is $\sum_{t=0}^{\infty} q_t[f(k_t)n_t - w_t n_t - \eta_t k_t n_t]$, where k_t is the firm's capital-labor ratio and n_t is the amount of labor it hires. The government sets g_t exogenously and must set τ_{ct}, τ_{ht} to satisfy the budget constraint:

$$(1) \quad \sum_{t=0}^{\infty} q_t(\tau_{ct}c_t + \tau_{ht}) = \sum_{t=0}^{\infty} q_t g_t.$$

- a. Define a competitive equilibrium.
- b. Suppose that historically the government had unlimited access to lump-sum taxes and availed itself of them. Thus, for a long time the economy had $g_t = \bar{g} > 0$, $\tau_{ct} = 0$. Suppose that this situation had been expected to go on forever. Tell how to find the steady-state capital-labor ratio for this economy.
- c. In the economy depicted in b, prove that the timing of lump-sum taxes is irrelevant.
- d. Let $\bar{k}_0$ be the steady value of k_t that you found in part b. Let this be the initial value of capital at time $t = 0$ and consider the following experiment. Suddenly and unexpectedly, a court decision rules that lump-sum taxes are illegal and that starting at time $t = 0$, the government must finance expenditures using the consumption tax τ_{ct} . The value of g_t remains constant at $\bar{g}$. Policy advisor number 1 proposes the following tax policy: find a *constant* consumption tax that satisfies the budget constraint (1), and impose it from time 0 onward. Please compute the new steady-state value of k_t under this policy. Also, get as far as you can in analyzing the transition path from the old steady state to the new one.
- e. Policy advisor number 2 proposes the following alternative policy. Instead of imposing the increase in τ_{ct} suddenly, he proposes to ease the pain by postponing the increase for 10 years. Thus, he/she proposes to set $\tau_{ct} = 0$ for $t = 0, \dots, 9$, then to set $\tau_{ct} = \bar{\tau}_c$ for $t \geq 10$. Please compute the steady-state level of capital associated with this policy. Can you say anything about the transition path to the new steady-state k_t under this policy?
- f. Which policy is better, the one recommended in d or the one in e?

Solution:

(a) A competitive equilibrium is a price system $\{q_t, \eta_t, w_t\}$, a government policy $\{g_t, \tau_{ct}, \tau_{ht}\}$ that satisfies (1), and an allocation $\{c_t, k_{t+1}, n_t\}$ such that: (i) given prices and taxes, $\{c_t, k_{t+1}\}$ maximises $\sum_t \beta^t u(c_t)$ subject to the household budget, with k_0 given; (ii) given prices, $\{k_t, n_t\}$ maximises the value of the firm; (iii) markets clear: $n_t = 1$, the firm's capital equals the household's, and feasibility holds with equality. Conditions. Collect the terms in k_{t+1} in the household budget (as in (11.5.1)): the coefficient of k_{t+1} is $-q_t + q_{t+1}(\eta_{t+1} + 1 - \delta)$. A bounded budget requires it to be zero, so

$$q_t = q_{t+1}(\eta_{t+1} + 1 - \delta). \quad (11.1.1)$$

FOC for c_t with budget multiplier μ : $\beta^t u'(c_t) = \mu q_t(1 + \tau_{ct})$. Firm (constant returns): $\eta_t = f'(k_t)$, $w_t = f(k_t) - k_t f'(k_t)$. Divide the FOC at t by the FOC at $t + 1$ and use (11.1.1):

$$\frac{u'(c_t)}{1 + \tau_{ct}} = \beta \frac{u'(c_{t+1})}{1 + \tau_{c,t+1}} [f'(k_{t+1}) + 1 - \delta], \quad c_t = f(k_t) + (1 - \delta)k_t - g_t - k_{t+1}, \quad (11.1.2)$$

plus k_0 given and the terminal condition $\lim_T \beta^T u'(c_T) k_{T+1} / (1 + \tau_{cT}) = 0$.

(b) With $\tau_{ct} = 0$ and $g_t = \bar{g}$, a steady state has $c_{t+1} = c_t$, so (11.1.2) gives $1 = \beta[f'(\bar{k}_0) + 1 - \delta]$:

$$f'(\bar{k}_0) = \rho + \delta, \quad \bar{c}_0 = f(\bar{k}_0) - \delta \bar{k}_0 - \bar{g}.$$

$\bar{g}$ does not enter the equation for $\bar{k}_0$. Lump-sum taxes adjust so that $\sum_t q_t \tau_{ht} = \sum_t q_t \bar{g}$.

(c) τ_{ht} enters the household's problem only through $\sum_t q_t \tau_{ht}$, and the government's only through the same sum. Take two tax sequences $\{\tau_{ht}\}, \{\tau'_{ht}\}$ with equal present value at the equilibrium prices. Then the household faces the same budget set, so it makes the same choices; the firm's problem does not involve taxes; the government budget still holds. Hence the same prices and allocation are an equilibrium under both sequences. Equivalently, the conditions (11.1.2) that determine the allocation contain no τ_h .

(d) Let $\tau_{ct} = \bar{\tau}$ for all $t \geq 0$. The ratio $(1 + \bar{\tau})/(1 + \bar{\tau}) = 1$ cancels from (11.1.2), which becomes the equation of part (b). Since $k_0 = \bar{k}_0$, the constant path $k_t = \bar{k}_0$, $c_t = \bar{c}_0$ satisfies the Euler equation, feasibility and the terminal condition. So

new steady state = $\bar{k}_0$, no transition: $k_t = \bar{k}_0$, $c_t = \bar{c}_0$ for all t .

The tax rate: with $\tau_{ht} = 0$ and $c_t = \bar{c}_0$, (1) reads $\bar{\tau} \bar{c}_0 \sum_t q_t = \bar{g} \sum_t q_t$, so $\bar{\tau} = \bar{g}/\bar{c}_0$. A constant τ_c with inelastic labour and no leisure is a tax on the present value of lifetime consumption, i.e. on wealth: nothing the household does can avoid it, so it works like a lump-sum tax. Only the level of q_t changes ($q_t = \beta^t u'(\bar{c}_0)/(\mu(1 + \bar{\tau}))$), not relative prices.

(e) *Steady state.* From $t = 10$ on τ_c is constant, so the steady-state version of (11.1.2) is again $f'(\bar{k}) = \rho + \delta$: the new steady state is $\bar{k}_0$.

Transition. For $t \neq 9$ the tax ratio in (11.1.2) is 1. At $t = 9$,

$$u'(c_9) = \beta u'(c_{10}) \frac{f'(k_{10}) + 1 - \delta}{1 + \bar{\tau}_c}.$$

The anticipated rise in τ_c acts like a one-time tax at rate $\bar{\tau}_c/(1 + \bar{\tau}_c)$ on the gross return of capital carried from $t = 9$ to $t = 10$: goods consumed before 10 are cheap relative to goods consumed after. Consequences (the same mechanism as figure 11.9.4):

- c jumps up at $t = 0$ and capital is run down between 0 and 10 (a consumption binge);
- as k falls, $f'(k)$ rises, so $\bar{R}_{t+1} = f'(k_{t+1}) + 1 - \delta > 1 + \rho$ for $t + 1 \leq 9$ and, by (11.1.2), c keeps rising until $t = 9$;
- at $t = 10$, $\bar{R}_{10} = [f'(k_{10}) + 1 - \delta]/(1 + \bar{\tau}_c) < 1$: c falls sharply;
- after $t = 10$ the economy is at $k_{10} < \bar{k}_0$ with constant policy: k and c rise monotonically back to $(\bar{k}_0, \bar{c}_0)$ and $\bar{R}$ falls to $1 + \rho$.

$\bar{\tau}_c$ solves a fixed point, because prices depend on it: $\bar{\tau}_c \sum_{t \geq 10} q_t c_t = \bar{g} \sum_{t \geq 0} q_t$ with $q_t = \beta^t u'(c_t)/(1 + \tau_{ct})$ from the equilibrium path computed at that $\bar{\tau}_c$.

(f) Policy (d) is better. With a constant τ_c the equilibrium conditions (11.1.2) coincide with the first-order conditions of a planner who takes $\{g_t\}$ as given and maximises

$\sum_t \beta^t u(c_t)$ subject to feasibility (section 11.8, item 1); the planner's problem is strictly concave, so (d) implements its unique solution. The allocation of (e) is feasible for the planner but violates the planner's Euler equation at $t = 9$; it therefore gives strictly lower utility. Postponing the tax does not “ease the pain”: it replaces a tax on wealth by a distorting tax on saving at one date.

Numerical exercise — see the companion code

Benchmark parameters. (d) $\bar{\tau} = \bar{g}/\bar{c}_0 = .2/.642645 = .311214$; the Newton path is exactly $k_t = 1.489956$. (e) $\bar{\tau}_c = .623899$ (found by `brentq` on the PV budget). $c_0 = .658407$, $c_9 = .702440$, $c_{10} = .564675$; $k_{10} = 1.126850$ is the minimum of k ; $\bar{R}_{10} = .680229$. Shooting reproduces the Newton path (to 10^{-8}). (f) Welfare -31.1214 (d) vs -31.2574 (e): (e) costs 0.435% of consumption every period.

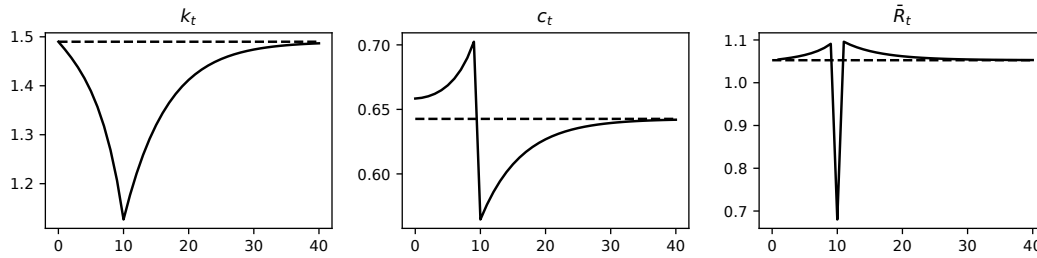


Figure 1: Exercise 11.1(e): consumption tax postponed to $t = 10$ (solid) vs constant tax from $t = 0$ (dashed).

Exercise 11.2 Tax reform: II

Question

p. 451. Consider the following economy populated by a government and a representative household. There is no uncertainty, and the economy and the representative household and government within it last forever. The government consumes a constant amount $g_t = g > 0$, $t \geq 0$. The government also sets sequences of two types of taxes, $\{\tau_{ct}, \tau_{kt}\}_{t=0}^{\infty}$. Here τ_{ct}, τ_{kt} are, respectively, a possibly time-varying flat-rate tax on consumption and a time-varying flat-rate tax on earnings from capital. The preferences of the household are ordered by $\sum_{t=0}^{\infty} \beta^t u(c_t)$, where $\beta \in (0, 1)$ and $u(\cdot)$ is strictly concave, increasing, and twice continuously differentiable. The household is endowed with one unit of labor each period and does not value leisure. The feasibility condition in the economy is $g_t + c_t + k_{t+1} \leq f(k_t) + (1 - \delta)k_t$ where k_t is the stock of capital owned by the household at the beginning of time t and $\delta \in (0, 1)$ is a depreciation rate. At time 0, there are complete markets for commodities at all dates. The household faces the budget constraint:

$$\sum_{t=0}^{\infty} \{q_t [(1 + \tau_{ct})c_t + k_{t+1} - (1 - \delta)k_t]\} \leq \sum_{t=0}^{\infty} q_t \{\eta_t k_t - \tau_{kt}(\eta_t - \delta)k_t + w_t\}$$

where we assume that the household inelastically supplies one unit of labor, and q_t is the price of date t consumption goods in units of the numeraire at time 0, η_t is the rental rate of date

t capital in units of time t goods, and w_t is the wage rate of date t labor in units of time t goods.

A production firm rents labor and capital. The value of the firm is $\sum_{t=0}^{\infty} q_t [f(k_t)n_t - w_t n_t - \eta_t k_t n_t]$, where here k_t is the firm's capital-labor ratio and n_t is the amount of labor it hires. The government sets $\{g_t\}$ exogenously and must set the sequences $\{\tau_{ct}, \tau_{kt}\}$ to satisfy the budget constraint:

$$(1) \quad \sum_{t=0}^{\infty} q_t (\tau_{ct} c_t + \tau_{kt} (\eta_t - \delta) k_t) = \sum_{t=0}^{\infty} q_t g_t.$$

a. Define a competitive equilibrium.

b. Assume an initial situation in which from time $t \geq 0$ onward, the government finances a constant stream of expenditures $g_t = \bar{g}$ entirely by levying a constant tax rate τ_k on capital and a zero consumption tax. Tell how to find steady-state levels of capital, consumption, and the rate of return on capital.

c. Let $\bar{k}_0$ be the steady value of k_t that you found in part **b**. Let this be the initial value of capital at time $t = 0$ and consider the following experiment. Suddenly and unexpectedly, a new party comes into power that repeals the tax on capital, sets $\tau_k = 0$ forever, and finances the same constant level of $\bar{g}$ with a flat-rate tax on consumption. Tell what happens to the new steady-state values of capital, consumption, and the return on capital.

d. Someone recommends comparing the two alternative policies of (1) relying completely on the taxation of capital as in the initial equilibrium and (2) relying completely on the consumption tax, as in our second equilibrium, by comparing the discounted utilities of consumption in steady state, i.e., by comparing $\frac{1}{1-\beta} u(\bar{c})$ in the two equilibria, where $\bar{c}$ is the steady-state value of consumption. Is this a good way to measure the costs or gains of one policy *vis-a-vis* the other?

Solution:

(a) As in 11.1(a), with the policy $\{g_t, \tau_{ct}, \tau_{kt}\}$ satisfying (1). The coefficient of k_{t+1} in the household budget is now $-q_t + q_{t+1}[(1 - \tau_{k,t+1})(\eta_{t+1} - \delta) + 1]$, so no arbitrage (11.5.2) gives $q_t/q_{t+1} = (1 - \tau_{k,t+1})(\eta_{t+1} - \delta) + 1$. With $\beta^t u'(c_t) = \mu q_t(1 + \tau_{ct})$ and $\eta = f'$, $w = f - k f'$, the equilibrium solves (11.6.3) and (11.6.1), k_0 given, and the terminal condition (11.5.3).

(b) In a steady state (11.6.3) gives $1 = \beta[(1 - \tau_k)(f'(\bar{k}) - \delta) + 1]$, i.e.

$$f'(\bar{k}) = \delta + \frac{\rho}{1 - \tau_k} \quad (11.6.7), \quad \bar{c} = f(\bar{k}) - \delta \bar{k} - \bar{g}.$$

Government budget: with everything constant, $q_t = \beta^t q_0$ and (1) becomes $\tau_k(f'(\bar{k}) - \delta)\bar{k} \sum_t q_t = \bar{g} \sum_t q_t$. Using (11.6.7), $(f'(\bar{k}) - \delta) = \rho/(1 - \tau_k)$, so the two unknowns $(\bar{k}, \tau_k)$ solve

$$\bar{k} = \bar{k}(\tau_k) \equiv f'^{-1}\left(\delta + \frac{\rho}{1 - \tau_k}\right), \quad \mathcal{R}(\tau_k) \equiv \frac{\tau_k \rho \bar{k}(\tau_k)}{1 - \tau_k} = \bar{g}.$$

$\mathcal{R}$ is a Laffer curve: zero at $\tau_k = 0$, rising, then falling as $\bar{k}(\tau_k) \rightarrow 0$. Take the smallest root. Returns: the after-tax gross return is $\bar{R} = 1 + \rho$ (11.6.10) whatever τ_k ; the pre-tax gross return $f'(\bar{k}) + 1 - \delta = 1 + \rho/(1 - \tau_k)$ exceeds it.

Numbers. With $f = k^{.33}$ the peak revenue is $\max \mathcal{R} = .146$ (at $\tau_k \approx .91$) $< .2$: the benchmark $g = .2$ cannot be financed by a capital tax alone. We use $\bar{g} = .05$: $\tau_k = .445443$, $\bar{k}_0 = 1.182709$, $\bar{c} = .770397$, pre-tax $f' + 1 - \delta = 1.094907$, $\bar{R} = 1.052632$.
(c) With $\tau_k = 0$ and a constant τ_c , the tax ratio cancels from (11.6.3), so the equilibrium is the planner's path from $\bar{k}_0$ with $g_t = \bar{g}$. New steady state: $f'(k^*) = \rho + \delta$. Since $\rho + \delta < \delta + \rho/(1 - \tau_k)$ and f' is decreasing, $k^* > \bar{k}_0$. Consumption $c^* = f(k^*) - \delta k^* - \bar{g} > \bar{c}$ because $f(k) - \delta k$ is increasing for k below the golden rule $f'(k) = \delta$, and k^* is below it. The after-tax return in steady state is $1 + \rho$ in both regimes; the pre-tax return falls from $1 + \rho/(1 - \tau_k)$ to $1 + \rho$. Along the transition k rises monotonically, $\bar{R}_t > 1 + \rho$ falls, and c rises; to build capital, consumption first falls below its old level. The constant tax is $\tau_c = \bar{g} \sum_t q_t / \sum_t q_t c_t$, computed on the transition path.

Numbers. $k^* = 1.489956$, $c^* = .792645$, pre-tax return 1.052632; $c_0 = .721969 < \bar{c} = .770397$; $\tau_c = .064569$ (above the steady-state ratio $\bar{g}/c^* = .063080$, because early consumption is low).

(d) No. Comparing $u(\bar{c})/(1 - \beta)$ in the two steady states ignores that the economy starts at $\bar{k}_0$, not at k^* . To reach k^* the household must first cut consumption and invest. The right comparison is $\sum_t \beta^t u(c_t)$ along the equilibrium path from $\bar{k}_0$ (part c) against $u(\bar{c}_0)/(1 - \beta)$. The steady-state comparison overstates the gain from the reform. The gain is still positive: the path of (c) is the planner's optimum from $\bar{k}_0$, and staying at $(\bar{k}_0, \bar{c}_0)$ is feasible, so the reform cannot lower welfare.

Numerical exercise — see the companion code

$\bar{g} = .05$. Consumption-equivalent gain of the reform: 2.888% comparing steady states, 0.595% including the transition. Newton and shooting agree to 10^{-8} . Figure: transition (left, centre) and the steady-state Laffer curve $\mathcal{R}(\tau_k)$ with the level $\bar{g}$ (right).

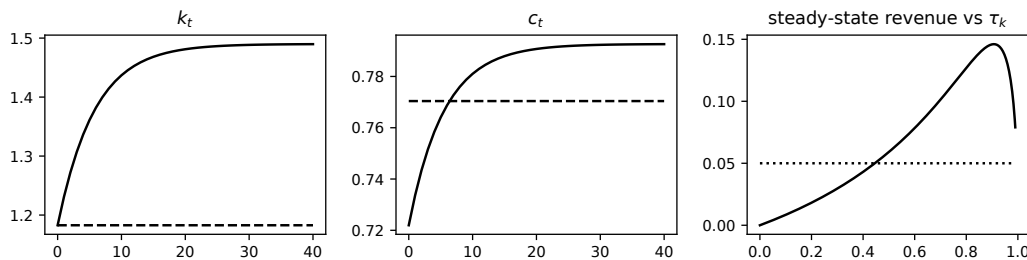


Figure 2: Exercise 11.2: transition after the capital-tax repeal (dashed: initial steady state).

Exercise 11.3 Anticipated productivity shift

Question

p. 453. An infinitely lived representative household has preferences over a stream of consumption of a single good that are ordered by $\sum_{t=0}^{\infty} \beta^t u(c_t)$, $\beta \in (0, 1)$, where u is a strictly concave, twice continuously differentiable, one-period utility function, β is a discount factor, and c_t is time t consumption. The technology is:

$$c_t + x_t \leq f(k_t)n_t, \quad k_{t+1} = (1 - \delta)k_t + \psi_t x_t$$

where for $t \geq 1$, $\psi_t = 1$ for $t < 4$ and $\psi_t = 2$ for $t \geq 4$. Here $f(k_t)n_t$ is output, where $f > 0$, $f' > 0$, $f'' < 0$, k_t is capital per unit of labor input, and n_t is labor input. The household supplies one unit of labor inelastically. The initial capital stock k_0 is given and is owned by the representative household. In particular, assume that k_0 is at the optimal steady value for k presuming that ψ_t had been equal to 1 forever. There is no uncertainty. There is no government.

- Formulate the planning problem for this economy in the space of sequences and form the pertinent Lagrangian. Find a formula for the optimal steady-state level of capital. How does a permanent increase in ψ affect the steady values of k , c , and x ?
- Formulate the planning problem for this economy recursively (i.e., compose a Bellman equation for the planner). Be careful to give a complete description of the state vector and its law of motion. (“Finding the state is an art.”)
- Formulate an (Arrow-Debreu) competitive equilibrium with time 0 trades, assuming the following decentralization. Let the household own the stocks of capital and labor and in each period let the household rent them to the firm. Let the household choose the investment rate each period. Define an appropriate price system and compute the first-order necessary conditions for the household and for the firm.
- What is the connection between a solution of the planning problem and the competitive equilibrium in part c? Please link the prices in part c to corresponding objects in the planning problem.
- Assume that k_0 is given by the steady-state value that corresponds to the assumption that ψ_t had been equal to 1 forever, and had been expected to remain equal to 1 forever. Qualitatively describe the evolution of the economy from time 0 on. Does the jump in ψ at $t = 4$ have any effects that precede it?

Solution: The statement defines ψ_t for $t \geq 1$; we take $\psi_0 = 1$. We do not impose $x_t \geq 0$ (it never binds in the computed path).

(a) With $n_t = 1$, eliminate $x_t = [k_{t+1} - (1 - \delta)k_t]/\psi_t$:

$$\max \sum_t \beta^t u(c_t) \quad \text{s.t.} \quad c_t + \frac{k_{t+1} - (1 - \delta)k_t}{\psi_t} \leq f(k_t), \quad k_0 \text{ given;}$$

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t) + \mu_t \left[f(k_t) - c_t - \frac{k_{t+1} - (1 - \delta)k_t}{\psi_t} \right] \right\}.$$

FOCs: c_t : $u'(c_t) = \mu_t$. k_{t+1} : $-\mu_t/\psi_t + \beta\mu_{t+1}[f'(k_{t+1}) + (1-\delta)/\psi_{t+1}] = 0$. Hence

$$\frac{u'(c_t)}{\psi_t} = \beta u'(c_{t+1}) \left[f'(k_{t+1}) + \frac{1-\delta}{\psi_{t+1}} \right], \quad \lim_{t \rightarrow \infty} \beta^t \frac{u'(c_t)}{\psi_t} k_{t+1} = 0. \quad (11.3.1)$$

Steady state with constant ψ : $1/\psi = \beta[f'(\bar{k}) + (1-\delta)/\psi]$, so $1 = \beta\psi f'(\bar{k}) + \beta(1-\delta)$ and

$$f'(\bar{k}) = \frac{\rho + \delta}{\psi}, \quad \bar{x} = \frac{\delta \bar{k}}{\psi}, \quad \bar{c} = f(\bar{k}) - \frac{\delta \bar{k}}{\psi}.$$

Comparative statics. $d\bar{k}/d\psi = -f'(\bar{k})/(\psi f''(\bar{k})) > 0$. $d\bar{c}/d\psi = [f'(\bar{k}) - \delta/\psi] d\bar{k}/d\psi + \delta\bar{k}/\psi^2 = (\rho/\psi) d\bar{k}/d\psi + \delta\bar{k}/\psi^2 > 0$. With the elasticity $\varepsilon \equiv (\psi/\bar{k}) d\bar{k}/d\psi = -f'(\bar{k})/(\bar{k} f'')$, $d\bar{x}/d\psi = (\delta/\psi) d\bar{k}/d\psi - \delta\bar{k}/\psi^2 = (\delta\bar{k}/\psi^2)(\varepsilon - 1)$: investment in goods rises iff $\varepsilon > 1$. For $f = zk^\alpha$, $\bar{k} \propto \psi^{1/(1-\alpha)}$ and $\varepsilon = 1/(1-\alpha) > 1$, so x rises. So a permanent rise in ψ raises k , c and (Cobb–Douglas) x .

(b) The problem is not stationary in t , but the only relevant information about the future is how far the economy is from the shift. State: (k_t, s_t) with $s_t = \min\{t, 4\}$, law of motion $s_{t+1} = \min\{s_t + 1, 4\}$, and $\psi(s) = 1$ for $s < 4$, $\psi(4) = 2$ (so $\psi_t = \psi(s_t)$). Bellman equation:

$$V(k, s) = \max_{k'} \left\{ u\left(f(k) - \frac{k' - (1-\delta)k}{\psi(s)}\right) + \beta V(k', \min\{s+1, 4\}) \right\}.$$

$V(\cdot, 4)$ is the stationary value function with $\psi = 2$ (a fixed point); $V(\cdot, 3), \dots, V(\cdot, 0)$ follow by four backward Bellman steps. FOC $u'(c)/\psi(s) = \beta V_k(k', s')$ and envelope $V_k(k, s) = u'(c)[f'(k) + (1-\delta)/\psi(s)]$ give (11.3.1).

(c) Price system $\{q_t, \eta_t, w_t\}$: q_t is the time-0 price of a unit of the time- t good (consumption and investment are the same good), η_t the rental rate of capital and w_t the wage, both in time- t goods. Household:

$$\max \sum_t \beta^t u(c_t) \quad \text{s.t.} \quad \sum_t q_t(c_t + x_t) \leq \sum_t q_t(\eta_t k_t + w_t), \quad k_{t+1} = (1-\delta)k_t + \psi_t x_t.$$

Substituting x_t , the coefficient of k_{t+1} in the budget is $-q_t/\psi_t + q_{t+1}[\eta_{t+1} + (1-\delta)/\psi_{t+1}]$. FOCs:

$$\beta^t u'(c_t) = \mu q_t, \quad \frac{q_t}{\psi_t} = q_{t+1} \left[\eta_{t+1} + \frac{1-\delta}{\psi_{t+1}} \right].$$

The price of a unit of installed capital is q_t/ψ_t (i.e. $1/\psi_t$ goods). Firm: $\max \sum_t q_t[f(k_t)n_t - w_t n_t - \eta_t k_t n_t]$ gives $\eta_t = f'(k_t)$, $w_t = f(k_t) - k_t f'(k_t)$.

(d) Substituting the firm's conditions and $q_t = \beta^t u'(c_t)/\mu$ into the household's conditions gives exactly (11.3.1) and feasibility: the competitive allocation solves the planning problem and conversely (welfare theorems). The prices are the planner's multipliers: $q_t = \beta^t \mu_t/\mu$ (resource constraint, discounted), q_t/ψ_t the shadow value of installed capital, $\eta_t = f'(k_t)$ its marginal product, w_t the marginal product of labour.

(e) Yes, the shift has effects before $t = 4$. Two forces act from $t = 0$:

- *Wealth.* Future output will be higher, so the household wants to consume more now.
- *Capital loss.* Capital bought at $t = 3$ costs one good per unit, but at $t = 4$ new capital costs only $1/2$ a good. By (11.3.1), the gross return from 3 to 4 is $\psi_3[f'(k_4) + (1 - \delta)/\psi_4] = f'(k_4) + (1 - \delta)/2$: the undepreciated capital loses half its value. Saving into $t = 4$ is unattractive.

So between $t = 0$ and 3 consumption rises and investment falls, and k declines. The low return from 3 to 4 implies $c_4 < c_3$ by (11.3.1). At $t = 4$ investment becomes twice as productive, investment surges, and k and c rise toward the new steady state, with high returns along the way.

Numerical exercise — see the companion code

$g = 0$, $\psi_0 = 1$. Steady states: $\psi = 1$: $(k, x, c) = (1.489956, .297991, .842645)$; $\psi = 2$: $(4.192491, .419249, 1.185532)$, with k multiplied by $2^{1/(1-\alpha)}$.

t	0	1	2	3	4	5	6	7
k_t	1.4900	1.4317	1.3672	1.2922	1.2009	1.5916	1.9555	2.2835
c_t	.9009	.9039	.9103	.9211	.7468	.8246	.8882	.9402
x_t	.2397	.2218	.1984	.1671	.3154	.3411	.3595	.3730
return $t \rightarrow t + 1$	1.0595	1.0676	1.0779	.6919	1.2834	1.2211	1.1796	1.1504

Newton and shooting agree to 10^{-8} ; $x_t > 0$ throughout.

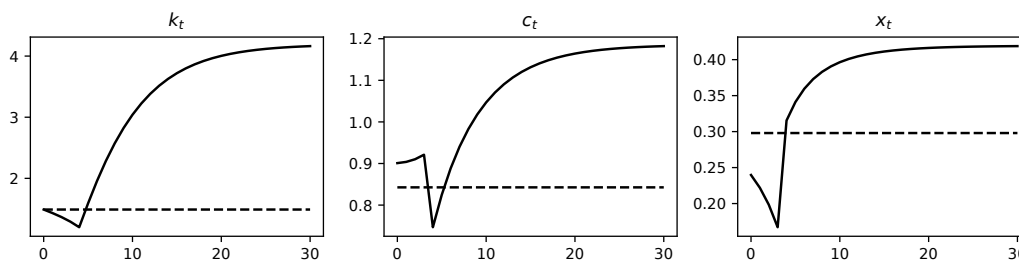


Figure 3: Exercise 11.3: responses to the shift in ψ at $t = 4$, known at $t = 0$ (dashed: old steady state).

Exercise 11.4 A capital levy

Question

p. 454. A nonstochastic economy produces one good that can be allocated among consumption, c_t , government purchases, g_t , and gross investment, x_t . The economy-wide resource constraints are

$$c_t + g_t + x_t \leq f(k_t), \quad k_{t+1} = (1 - \delta)k_t + x_t,$$

where $\delta \in (0, 1)$ is a depreciation rate, k_t is the capital stock, and $f(k_t)$ gives production as a function of capital, where $f(k) = Ak^\alpha$ with $\alpha \in (0, 1)$. A single representative consumer owns

the capital stock and one unit of labor. The consumer rents capital and labor to a competitive firm each period. The consumer ranks consumption plans according to $\sum_{t=0}^{\infty} \beta^t u(c_t)$ where $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, with $\gamma \geq 1$. The household supplies one unit of labor inelastically each period. The government has only one tax at its disposal, a one-time capital levy through which it confiscates part of the capital stock from the private sector. When the government imposes a capital levy, we assume that it sends the consumer a tax bill for a fraction ϕ of the beginning of period capital stock. Below, you will be asked to compare consequences of levying this tax either at the beginning of time $T = 0$ or at the beginning of time $T = 10$. The fraction can exceed 1 if that is necessary to finance the government budget. The government is allowed to impose no other taxes. Among other things, this means that it cannot impose a direct lump sum or ‘head’ tax.

- a. Define a competitive equilibrium with time 0 trading.
- b. Suppose that before time 0 the economy had been in a steady state in which g had always been zero and had been expected always to equal zero. Find a formula for the initial steady state capital stock in a competitive equilibrium with time zero trading. Let this value be $\bar{k}_0$.
- c. At time 0, everyone suddenly wakes up to discover that from time 0 on, government expenditures will be $\bar{g} > 0$, where $\bar{g} + \delta \bar{k}_0 < f(\bar{k}_0)$, which implies that the new level of government expenditures would be feasible in the old steady state. Suppose that the government finances the new path of expenditures by a capital levy at time $T = 0$. The government imposes a capital levy by sending the household a bill for a fraction of the value of its capital at the time indicated. Find the new steady state value of the capital stock in a competitive equilibrium. Describe an algorithm to compute the fraction of the capital stock that the government must tax away at time 0 to finance its budget. Find the new steady state value of the capital stock in a competitive equilibrium. Describe the time paths of capital, consumption, and the interest rate from $t = 0$ to $t = +\infty$ in the new equilibrium and compare them with their counterparts in the initial $g_t \equiv 0$ equilibrium.
- d. Assume the same new path of government expenditures indicated in part c, but now assume that the government imposes the one-time capital levy at time $T = 10$, and that this is foreseen at time 0. Find the new steady state value of the capital stock in a competitive equilibrium that is associated with this tax policy. Describe an algorithm to compute the fraction of the capital stock that the government must tax away at time $T = 10$ to finance its budget. Describe the time paths of capital, consumption, and the interest rate in this new equilibrium and compare them with their counterparts in part b and in the initial $g_t \equiv 0$ equilibrium.
- e. Define a competitive equilibrium with sequential trading of one-period Arrow securities. Describe how to compute such an equilibrium. Describe the time path of the consumer’s holdings of one-period securities in a competitive equilibrium with one period Arrow securities under the government tax policy assumed in part d. Describe the time path of government debt.

Solution: *Reading.* The bill is ϕk_T units of time- T goods; the household keeps its capital and its rentals. (Confiscating ϕk_T physical units is the same thing, since capital and goods convert one-for-one.) Parts (d) and (e) compare with “part b”, which we read as the initial $g \equiv 0$ steady state.

(a) A competitive equilibrium is prices $\{q_t, \eta_t, w_t\}$, a policy $(\{g_t\}, \phi, T)$, and an alloca-

tion $\{c_t, x_t, k_{t+1}\}$ such that: the household maximises $\sum_t \beta^t u(c_t)$ subject to

$$\sum_t q_t(c_t + x_t) \leq \sum_t q_t(\eta_t k_t + w_t) - q_T \phi k_T, \quad k_{t+1} = (1 - \delta)k_t + x_t, \quad k_0 \text{ given};$$

the firm sets $\eta_t = f'(k_t) = \alpha A k_t^{\alpha-1}$, $w_t = f(k_t) - k_t f'(k_t)$; the government budget holds, $q_T \phi k_T = \sum_t q_t g_t$; and $c_t + g_t + x_t = f(k_t)$. Collecting terms in k_{t+1} gives no-arbitrage conditions $q_t = q_{t+1}[\eta_{t+1} + 1 - \delta]$ for $t + 1 \neq T$ and $q_{T-1} = q_T[\eta_T + 1 - \delta - \phi]$ if $T \geq 1$. With $\beta^t u'(c_t) = \mu q_t$:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta - \phi \mathbf{1}\{t + 1 = T\}]. \quad (11.4.1)$$

(b) $g \equiv 0$, steady state: $1 = \beta[f'(\bar{k}_0) + 1 - \delta]$, so $\alpha A \bar{k}_0^{\alpha-1} = \rho + \delta$ and $\bar{k}_0 = (\alpha A / (\rho + \delta))^{1/(1-\alpha)}$; $\bar{c}_0 = A \bar{k}_0^\alpha - \delta \bar{k}_0$.

(c) $T = 0$: k_0 is given, so the bill ϕk_0 cannot be avoided and is a lump-sum tax; ϕ never enters (11.4.1) (which involves k_{t+1} , $t + 1 \geq 1$). The allocation is the planner's with $g_t = \bar{g}$. New steady state: from (11.4.1), $f'(\bar{k}) = \rho + \delta$, so $\bar{k} = \bar{k}_0$ (the level of g does not enter). Since $k_0 = \bar{k}_0$ is already the new steady state, the equilibrium is

$$k_t = \bar{k}_0, \quad c_t = \bar{c}_0 - \bar{g}, \quad q_t = \beta^t, \quad R_{t,t+1} = 1/\beta = 1 + \rho \quad \text{for all } t \geq 0.$$

Algorithm (general k_0 and $\{g_t\}$): (1) compute the planner's path by shooting on (11.4.1) with $\phi = 0$; (2) $q_t = \beta^t u'(c_t)/u'(c_0)$; (3) $\phi = \sum_t q_t g_t / (q_0 k_0)$. Here $\phi = \bar{g} / ((1 - \beta)\bar{k}_0)$, which exceeds 1 when $\bar{g} > (1 - \beta)\bar{k}_0$; the household pays out of its total wealth, including the present value of wages. *Comparison with $g \equiv 0$* : same capital and interest rate; consumption lower by exactly $\bar{g}$ at every date.

(d) $T = 10$ foreseen. Now k_{10} is chosen knowing the levy: by (11.4.1) the gross return on capital carried from 9 to 10 is $f'(k_{10}) + 1 - \delta - \phi$. The levy is a one-time tax on capital, a distorting tax. After $t = 10$ nothing distorts: the new steady state is again $\bar{k} = \bar{k}_0$.

Algorithm for ϕ . (1) Guess ϕ . (2) Solve (11.4.1) and feasibility with $g_t = \bar{g}$ by shooting from $\bar{k}_0$ toward k_{10} . (3) Compute $q_t = \beta^t u'(c_t)/u'(c_0)$. (4) Evaluate the gap $G(\phi) = \phi q_{10} k_{10} - \bar{g} \sum_t q_t$. (5) Update ϕ by bisection on the rising side of the Laffer curve. A Laffer curve appears because a higher ϕ lowers k_{10} : the Euler equation needs $f'(k_{10}) \approx \rho + \delta + \phi$. So revenue $\phi q_{10} k_{10}$ has a maximum, and an equilibrium exists only if that maximum covers $\bar{g} \sum_t q_t$.

Paths. Before 10, capital is run down (it will be taxed); consumption jumps above its $T = 0$ level and rises until $t = 9$, because $\bar{R}_{t+1} = f'(k_{t+1}) + 1 - \delta > 1 + \rho$ as k falls. The interest rate from 9 to 10 is very low (the return net of the levy). At 10 consumption drops. After 10 capital is rebuilt: the interest rate jumps above ρ and falls back, and consumption rises back to $\bar{c}_0 - \bar{g}$. Compared with the initial $g \equiv 0$ steady state: the long run is the same except that c is lower by $\bar{g}$. Compared with $T = 0$: the same long run, but a distorted transition and lower welfare.

Numerical exercise — see the companion code

$A = 1$, benchmark $\alpha, \beta, \delta, \gamma$: $\bar{k}_0 = 1.489956$, $\bar{c}_0 = .842645$. *Existence at $T = 10$* : the levy can finance $\bar{g}$ only if $\bar{g} \leq .03541$. With the benchmark $\bar{g} = .2$ the peak of the Laffer curve ($\phi = 5.80$) raises only 32% of the present value of spending, so no equilibrium exists. We use $\bar{g} = .02$.

(c) $\phi = \bar{g}/((1 - \beta)\bar{k}_0) = .268464$; $c_t = .822645$; $r = -\log \beta = .051293$. (d) $\phi = .494143$ (peak of the Laffer curve at $\phi = 1.516$); $k_{10} = 1.012095$; $c_0 = .840408$, $c_9 = .911460$, $c_{10} = .706924$; $r_{9,10} = -.456955$, $r_{10,11} = .106263$. Welfare -24.3118 ($T = 0$) > -24.4418 ($T = 10$). Newton and shooting agree to 10^{-8} .

(e) With no uncertainty, one-period Arrow securities are one-period risk-free bonds. Let b_t be the household's claims on time- t goods bought at $t - 1$ at price $1/R_{t-1,t}$, $b_0 = 0$. Household at each t :

$$c_t + k_{t+1} + \frac{b_{t+1}}{R_{t,t+1}} = \eta_t k_t + w_t + (1 - \delta)k_t + b_t - \phi k_t \mathbf{1}\{t = T\},$$

with a natural debt limit on b_{t+1} . Government: $g_t + b_t = b_{t+1}/R_{t,t+1} + \phi k_t \mathbf{1}\{t = T\}$. A sequential equilibrium is prices $\{R_{t,t+1}, \eta_t, w_t\}$ and quantities such that both budgets hold, the household and firm optimise, and the goods and bond markets clear. *Computation*: solve the time-0 equilibrium of (d); set $R_{t,t+1} = q_t/q_{t+1}$; then the bond position is the government debt in (11.4.3),

$$B_t \equiv \frac{b_{t+1}}{R_{t,t+1}} = \sum_{s>t} \frac{q_s}{q_t} (T_s - g_s), \quad T_s = \phi k_{10} \mathbf{1}\{s = 10\}.$$

Paths under (d). For $t < 10$ the government has no revenue and borrows: B_t grows ($B_t = g_t + R_{t-1,t}B_{t-1}$), and the household lends. At $t = 10$ the levy exceeds the debt and the government becomes a creditor: $B_{10} < 0$, i.e. the household is now in debt to the government. From then on the household pays interest on that debt, and the interest finances $\bar{g}$. In the limit $\bar{g} + RB = B$, so $B_\infty = -\bar{g}/\rho$.

Numerical exercise — see the companion code

$\bar{g} = .02$: $B_0 = .02$, $B_9 = .287525$, $B_{10} = -.298057$, $B_t \rightarrow -\bar{g}/\rho = -.38$. The sequential budget (11.4.2) $g_t + R_{t-1,t}B_{t-1} = T_t + B_t$ holds date by date (checked).

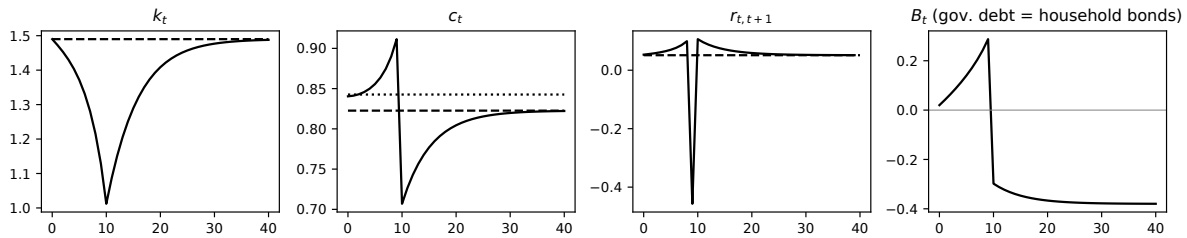


Figure 4: Exercise 11.4 ($\bar{g} = .02$): levy at $T = 10$ (solid) vs $T = 0$ (dashed); dotted: $g \equiv 0$ consumption.

Steady States, Open Economies and Linear Utility (11.5–11.9)

Exercise 11.5 Log utility, the golden rule and a jump in g *Question*

p. 456. A representative consumer has preferences ordered by $\sum_{t=0}^{\infty} \beta^t \log(c_t)$, $0 < \beta < 1$, where $\beta = \frac{1}{1+\rho}$, $\rho > 0$, and c_t is the consumption per worker. The technology is

$$y_t = f(k_t) = zk_t^\alpha, \quad 0 < \alpha < 1, \quad z > 0$$

where y_t is the output per unit labor and k_t is capital per unit labor

$$y_t = c_t + x_t + g_t, \quad k_{t+1} = (1 - \delta)k_t + x_t, \quad 0 < \delta < 1$$

and x_t is gross investment per unit of labor and g_t is government expenditures per unit of labor. Assume a competitive equilibrium with a price system $\{q_t, \eta_t, w_t\}_{t=0}^{\infty}$ and a government policy $\{g_t, \tau_{ht}\}_{t=0}^{\infty}$.

Assume that the government finances its expenditures by levying lump sum taxes. There are no distorting taxes. Assume that at time 0, the economy begins with a capital per unit of labor k_0 that equals the steady state value appropriate for an economy in which g_t had been zero forever.

- Find a formula for the steady state capital stock when $g_t = 0 \forall t$.
- Compare the steady state capital labor ratio $\bar{k}$ in the competitive equilibrium with the capital labor ratio $\tilde{k}$ that maximizes steady state consumption per capita, i.e., $\tilde{k}$ solves $\tilde{c} = \max_k (f(k) - \delta k)$. Is $\tilde{k}$ greater than or less than $\bar{k}$? If they differ, please tell why. Is $\tilde{c}$ greater or less than $\bar{c} = f(\bar{k}) - \delta \bar{k}$? Explain why.
- Now assume that at time 0, g_t suddenly jumps to the value $g = \frac{1}{2}\bar{c}$ where $\bar{c}$ is the value of consumption per capita in the initial steady state in which g was zero forever. Starting from $k_0 = \bar{k}$ for the old $g = 0$ steady state, find the time paths of $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ associated with the new path $g_t = g > 0 \forall t$ for government expenditures per capita. Also show the time path for $\bar{R}_{t+1} \equiv (1 - \delta) + f'(k_{t+1})$. Explain why the new paths are as they are.

Solution: With lump-sum taxes the equilibrium is the planner's allocation with g_t deducted from output (section 11.8), and (11.6.3) with $u = \log$ reads $c_{t+1}/c_t = \beta[f'(k_{t+1}) + 1 - \delta]$.

(a) Steady state: $1 = \beta[f'(\bar{k}) + 1 - \delta]$, i.e. $\alpha z \bar{k}^{\alpha-1} = \rho + \delta$:

$$\bar{k} = \left(\frac{\alpha z}{\rho + \delta} \right)^{\frac{1}{1-\alpha}}, \quad \bar{c} = z \bar{k}^\alpha - \delta \bar{k}.$$

(b) $\tilde{k}$ maximises $f(k) - \delta k$: $f'(\tilde{k}) = \delta$, so $\tilde{k} = (\alpha z / \delta)^{1/(1-\alpha)}$. Since $\rho + \delta > \delta$ and f' is decreasing, $\tilde{k} > \bar{k}$. And $\tilde{c} \geq \bar{c}$ by definition of $\tilde{k}$, strictly because $\bar{k} \neq \tilde{k}$ and $f(k) - \delta k$

is strictly concave. Why they differ: the household discounts the future. At $\bar{k}$ the net marginal product $f'(\bar{k}) - \delta = \rho$ just compensates for impatience. Moving from $\bar{k}$ toward $\tilde{k}$ would require giving up consumption now for a permanent gain later that yields less than ρ per period, so it is not worth it. $\bar{k}$ is the “modified” (or augmented, section 11.6.2) golden rule.

(c) g does not enter the steady-state Euler equation, so the new steady state is still $\bar{k}$. Check that the constant path

$$k_t = \bar{k}, \quad c_t = f(\bar{k}) - \delta\bar{k} - g = \bar{c} - g = \frac{1}{2}\bar{c}, \quad \bar{R}_{t+1} = f'(\bar{k}) + 1 - \delta = 1 + \rho \quad \forall t \geq 0$$

is the equilibrium: feasibility holds; $c_{t+1}/c_t = 1 = \beta(1 + \rho)$ is the Euler equation; the terminal condition holds. The solution of the (concave) planning problem is unique, so this is the only equilibrium. Explanation: a permanent, unforeseen rise in g financed by lump-sum taxes is a pure wealth effect. The household’s present value of taxes rises by $g \sum_t q_t$; it smooths the loss by cutting consumption by g at every date. The return on capital is unchanged, so there is no reason to change the capital stock. Government purchases crowd out consumption one-for-one and investment not at all.

Numerical exercise — see the companion code

$z = 1$, benchmark α, β, δ : $\bar{k} = 1.489956$, $\bar{c} = .842645$; $\tilde{k} = 2.111557$, $\tilde{c} = .857420$.
(c) $g = .421323$: the Newton path is $k_t = 1.489956$, $c_t = .421323$, $\bar{R}_{t+1} = 1.052632$ for all t (checked).

Exercise 11.6 Trade and growth

Question

p. 457. **Part I.** Consider the problem of a planner in a small economy. When the economy is closed to international trade, the planner chooses $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ to maximize $\sum_{t=0}^{\infty} \beta^t u(c_t)$, where $0 < \beta < 1$, $\beta = \frac{1}{1+\rho}$, $\rho > 0$ subject to

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t, \quad \delta \in (0, 1)$$

where $u(c_t) = \frac{c_t^{1-\gamma}}{1-\gamma}$, $\gamma > 0$, $f(k_t) = zk_t^\alpha$, and $0 < \alpha < 1$. Let $\bar{k}$ be the steady state value of k_t under the optimal plan.

- a. Find a formula for $\bar{k}$.
- b. Assume that $k_0 < \bar{k}$. Describe time paths for $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ and $\bar{R}_{t+1} = (1 - \delta) + f'(k_{t+1})$.
- c. What is the steady state value of $\bar{R}_{t+1}$?
- d. Is $\bar{R}_{t+1}$ less or greater than its steady state value when $k_{t+1} < \bar{k}$?

Part II. Now assume that the economy is open to international trade in capital and financial assets. Assume that there is a fixed world gross rate of return $R = \beta^{-1}$ at which the planner can borrow or lend, what is often called a ‘small open economy’ assumption. The planner can use the proceeds of borrowing to purchase goods on the international market. These goods can be used to augment capital or to consume. Let $\bar{k}_0$ be the level of initial capital ($\bar{k}_0 < \bar{k}$) before the country opens up to trade just before time 0. Let $\bar{k}_0$ be the same initial capital per capita $\bar{k}_0 < \bar{k}$ studied in parts a-d.

At time $t = -1$, after $\bar{k}_0$ was set, trade opens up. At time $t = -1$, the planner can issue IOU's or bonds in amount B_{-1} and use the proceeds to purchase capital, thereby setting $k_0 = \bar{k}_0 + B_{-1}$ where B_{-1} is denominated in time -1 consumption goods. The bonds are one period in duration and bear the constant world gross interest rate $R = \beta^{-1}$. For $t \geq 0$, the planner faces the constraints

$$c_t + k_{t+1} + RB_{t-1} = f(k_t) + (1 - \delta)k_t + B_t$$

Here RB_{t-1} is the sum of interest and principal on the bonds issued at $t - 1$ and B_t is the amount of one-period bonds issued at t .

The planning problem in the small open economy is now to choose $\{c_t, k_t, B_t\}_{t=0}^{\infty}$ and B_{-1} , subject to $\bar{k}_0$ given. (Notice that k_0 is a choice variable and that $\bar{k}_0$ is an initial condition.) Please solve the planning problem in the small open economy and compare outcomes to those in the closed economy. Is welfare $\sum_{t=0}^{\infty} \beta^t u(c_t)$ higher in the “open” or “closed” economy?

Solution:

Part I. (a) Lagrangian $\sum_t \beta^t \{u(c_t) + \mu_t [f(k_t) + (1 - \delta)k_t - c_t - k_{t+1}]\}$; FOCs $u'(c_t) = \mu_t$

and $\mu_t = \beta\mu_{t+1}[f'(k_{t+1}) + 1 - \delta]$, so $c_t^{-\gamma} = \beta c_{t+1}^{-\gamma} \bar{R}_{t+1}$. Steady state $\beta \bar{R} = 1$:

$$\alpha z \bar{k}^{\alpha-1} = \rho + \delta \quad \Rightarrow \quad \bar{k} = \left(\frac{\alpha z}{\rho + \delta} \right)^{\frac{1}{1-\alpha}}.$$

(b) With $k_0 < \bar{k}$, the optimal policy $k_{t+1} = h(k_t)$ is increasing with $h(k) > k$ for $k < \bar{k}$, so k_t rises monotonically to $\bar{k}$. Along the path $k_{t+1} < \bar{k}$, so $f'(k_{t+1}) > \rho + \delta$ and $\bar{R}_{t+1} > 1 + \rho$; then by the Euler equation $c_{t+1}/c_t = (\beta \bar{R}_{t+1})^{1/\gamma} > 1$: consumption rises monotonically to $\bar{c} = f(\bar{k}) - \delta \bar{k}$. $\bar{R}_{t+1}$ falls monotonically to $1 + \rho$ as k_{t+1} rises.

(c) $\bar{R} = f'(\bar{k}) + 1 - \delta = 1 + \rho = \beta^{-1}$.

(d) Greater: f' is decreasing, so $k_{t+1} < \bar{k}$ implies $f'(k_{t+1}) > f'(\bar{k})$ and $\bar{R}_{t+1} > 1 + \rho$.

Part II. Lagrangian with multipliers $\beta^t \mu_t$ on the period- t constraints ($t \geq 0$), using $k_0 = \bar{k}_0 + B_{-1}$:

$$\mathcal{L} = \sum_{t \geq 0} \beta^t \left\{ u(c_t) + \mu_t [f(k_t) + (1 - \delta)k_t + B_t - c_t - k_{t+1} - RB_{t-1}] \right\}.$$

FOCs:

- c_t : $u'(c_t) = \mu_t$.
- B_t ($t \geq 0$): $\beta^t \mu_t - \beta^{t+1} R \mu_{t+1} = 0$, so $u'(c_t) = \beta R u'(c_{t+1}) = u'(c_{t+1})$: *constant consumption*.
- k_{t+1} ($t \geq 0$): $\mu_t = \beta \mu_{t+1} [f'(k_{t+1}) + 1 - \delta]$, so $f'(k_{t+1}) + 1 - \delta = R = 1 + \rho$: $k_{t+1} = \bar{k}$.
- B_{-1} (enters through $k_0 = \bar{k}_0 + B_{-1}$ and the repayment RB_{-1}): $\mu_0 [f'(k_0) + 1 - \delta - R] = 0$, so $k_0 = \bar{k}$ and $B_{-1} = \bar{k} - \bar{k}_0 > 0$: the country borrows abroad to install the steady-state capital at once.

Consumption level. Multiply the constraint at t by $R^{-t} = \beta^t$ and sum; with $k_t = \bar{k}$ and the no-Ponzi condition $\lim_T R^{-T} B_T = 0$ the B terms telescope to $-RB_{-1}$:

$$\sum_t \beta^t c_t = \frac{f(\bar{k}) - \delta \bar{k}}{1 - \beta} - RB_{-1} \Rightarrow \boxed{c = f(\bar{k}) - \delta \bar{k} - \frac{1 - \beta}{\beta} B_{-1} = f(\bar{k}) - \delta \bar{k} - \rho(\bar{k} - \bar{k}_0)}.$$

Debt. From the constraint, $B_t - RB_{t-1} = c - f(\bar{k}) + \delta \bar{k} = -\rho B_{-1}$. At $t = 0$, $B_0 = (R - \rho)B_{-1} = B_{-1}$; by induction $B_t = B_{-1}$ for all t : the country rolls over its debt and pays interest ρB_{-1} each period.

Comparison. The open economy jumps immediately to $\bar{k}$ and has flat consumption and $\bar{R} = 1 + \rho$ throughout. The closed economy approaches $\bar{k}$ gradually, with rising consumption and $\bar{R} > 1 + \rho$. Welfare is higher in the open economy. The closed-economy plan ($B_t = 0$ for all t , $k_0 = \bar{k}_0$) is feasible in the open economy, so the open optimum is at least as good. It is strictly better because the closed plan has rising consumption, which violates the open economy's FOC $u'(c_t) = u'(c_{t+1})$.

Numerical exercise — see the companion code

$z = 1$, $g = 0$, $\bar{k}_0 = .5\bar{k}$. Closed: $k_0 = .744978 \rightarrow 1.489956$, $c_0 = .647540 \rightarrow .842645$, $\bar{R}_1 = 1.166272 \rightarrow 1.052632$ (monotone paths checked). Open: $B_{-1} = .744978$, $c = .803436$, interest $\rho B_{-1} = .039209$ per period; PV and period budgets checked. Welfare -25.2394 (closed) < -24.8931 (open): the open economy is worth 1.391% more consumption every period.

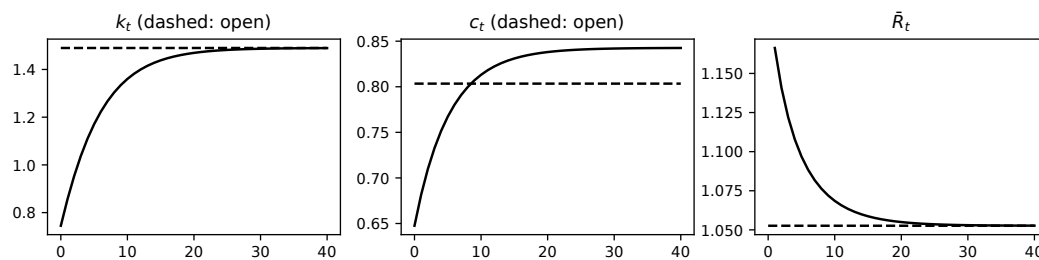


Figure 5: Exercise 11.6: closed economy from $\bar{k}_0 = .5\bar{k}$ (solid) and the small open economy (dashed).

Exercise 11.7 Capital taxes: unforeseen, foreseen, and the role of γ

Question

p. 458. The next several problems assume the following environment. A representative consumer has preferences ordered by $\sum_{t=0}^{\infty} \beta^t u(c_t)$, $0 < \beta < 1$, $\beta = \frac{1}{1+\rho}$, $\rho > 0$, where c_t is the consumption per worker and where $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ for $\gamma > 0$ and $\gamma \neq 1$, $u(c) = \log(c)$ if $\gamma = 1$. The technology is $y_t = f(k_t) = zk_t^\alpha$, $0 < \alpha < 1$, $z > 0$, where y_t is the output per unit labor and k_t is capital per unit labor and

$$y_t = c_t + x_t + g_t, \quad k_{t+1} = (1 - \delta)k_t + x_t, \quad 0 < \delta < 1$$

where x_t is gross investment per unit of labor and g_t is government expenditures per unit of labor. The government finances its expenditures by levying some combination of a flat rate tax τ_{ct} on the value of consumption goods purchased at t , a flat rate tax τ_{nt} on the value of labor earnings at t , a flat rate tax τ_{kt} on earnings from capital at t and a lump sum tax of τ_{ht} in time t consumption goods per worker at time t . Let $\{q_t, \eta_t, w_t\}_{t=0}^{\infty}$ be a price system.

Consider an economy in which $g_t = \bar{g} > 0 \forall t \geq 0$ and in which initially the government finances all expenditures by lump sum taxes.

a. Find a formula for the steady state capital labor ratio k_t for this economy. Find formulas for the steady state level of c_t and $\bar{R}_t = [(1 - \delta + f'(k_{t+1}))]$

b. Now suppose that starting from $k_0 = \bar{k}$, i.e., the steady state that you computed in part **a**, the government suddenly increases the tax on earnings from capital to a constant level $\tau_k > 0$. The government adjusts lump sum taxes to keep the government budget balances. Describe competitive equilibrium time paths for c_t , k_{t+1} , $\bar{R}_t$ and their relationship to corresponding values in the old steady state that you described in part **a**.

c. Describe how the shapes of the paths that you found in part **b** depend on the curvature parameter γ in the utility function $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$. Higher values of γ imply higher curvature and more aversion to consumption path that fluctuate. Higher values of γ imply that the consumer values smooth consumption paths even more.

d. Starting from the steady state $\bar{k}$ that you computed in part **a**, now consider a situation in which the government announces at time 0 that starting in period 10 the tax on earnings from capital τ_k will rise permanently to $\bar{\tau}_k > 0$. The government adjusts its lump sum taxes to balance its budget.

i) Find the new steady state values for $k_t, c_t, \bar{R}_t$.

ii) Describe the shapes of the transition paths from the initial steady states to the new one for $k_t, c_t, \bar{R}_t$.

iii) Describe how the shapes of the transition paths depend on the curvature parameter γ in the utility function $u(c)$.

Hint: When γ is bigger, consumers more strongly prefer smoother consumption paths. Recall the forces behind formula (11.10.16) in section 11.10.6.

Solution: Throughout, $\bar{R}_{t+1} = 1 + (1 - \tau_{k,t+1})(f'(k_{t+1}) - \delta)$ (no consumption tax) and (11.6.3) is $(c_{t+1}/c_t)^\gamma = \beta \bar{R}_{t+1}$. Lump-sum taxes absorb the budget, so they do not affect the allocation.

(a) $\beta \bar{R} = 1$ with $\tau_k = 0$:

$$\bar{k} = \left(\frac{\alpha z}{\rho + \delta} \right)^{\frac{1}{1-\alpha}}, \quad \bar{c} = z\bar{k}^\alpha - \delta\bar{k} - \bar{g}, \quad \bar{R} = 1 + \rho.$$

(b) New steady state from (11.6.7): $f'(\hat{k}) = \delta + \rho/(1 - \tau_k)$, so $\hat{k} = (\alpha z/(\delta + \rho/(1 - \tau_k)))^{1/(1-\alpha)} < \bar{k}$; $\hat{c} = f(\hat{k}) - \delta\hat{k} - \bar{g} < \bar{c}$ (since $\hat{k} < \bar{k} < \text{golden rule}$); after-tax $\bar{R} = 1 + \rho$ again.

Transition. At $t = 0$ the capital stock $k_0 = \bar{k}$ exceeds $\hat{k}$. At $\bar{k}$ the after-tax return is $1 + (1 - \tau_k)\rho < 1 + \rho$. As long as $k_{t+1} > \hat{k}$, $f'(k_{t+1}) < f'(\hat{k})$, so $\bar{R}_{t+1} < 1 + \rho$ and consumption falls: $c_{t+1} < c_t$. Hence:

- c jumps up at $t = 0$: $c_0 = f(\bar{k}) + (1 - \delta)\bar{k} - \bar{g} - k_1 > \bar{c}$ because $k_1 < \bar{k}$ (the household disinvests);
- then c_t and k_t decline monotonically to $\hat{c} < \bar{c}$ and $\hat{k} < \bar{k}$;
- $\bar{R}_t$ drops below $1 + \rho$ at $t = 1$ and rises monotonically back to $1 + \rho$.

(c) By (11.6.9), $\log(c_{t+1}/c_t) = \gamma^{-1} \log(\beta \bar{R}_{t+1})$. For a given gap between $\bar{R}$ and $1 + \rho$, a small γ (high elasticity of intertemporal substitution $1/\gamma$) produces large consumption growth rates. So consumption starts high and falls fast, capital is run down quickly, and $\bar{R}$ returns quickly to $1 + \rho$. A large γ gives a small initial jump in c and a slow, drawn-out decline of c and k . The speed is measured by the feedback coefficient λ_2 of (11.10.8): $k_{t+1} - \hat{k} \approx \lambda_2(k_t - \hat{k})$, and λ_2 increases with γ .

Deriving $\lambda_2(\gamma)$ and correcting (11.10.16). Linearise $u'(c_t) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]$ with $c_t = f(k_t) + (1 - \delta)k_t - g - k_{t+1}$ around a steady state without distortions ($\beta \bar{R} = 1$):

$$u'' dc_t = u'' dc_{t+1} + \beta u' f'' dk_{t+1} \Rightarrow dc_t = dc_{t+1} + \frac{A}{\gamma} dk_{t+1}, \quad A \equiv \beta \bar{c} |f''(\bar{k})|,$$

using $u'/u'' = -\bar{c}/\gamma$. With $dc_t = \beta^{-1} dk_t - dk_{t+1}$:

$$dk_{t+2} - b dk_{t+1} + \beta^{-1} dk_t = 0, \quad b = 1 + \frac{1}{\beta} + \frac{A}{\gamma}.$$

The roots of $\lambda^2 - b\lambda + \beta^{-1} = 0$ have product $1/\beta$ (as in section 11.10.1) and sum $b > 1 + 1/\beta$, so one root is below 1 and the other above $1/\beta$. The stable root is

$$\lambda_2 = \frac{b - \sqrt{b^2 - 4/\beta}}{2} = \frac{2/\beta}{b + \sqrt{b^2 - 4/\beta}} = \frac{\gamma}{D},$$

$$D = \frac{1 + \beta}{2} \gamma + \frac{\beta A}{2} + \frac{\beta}{2} \sqrt{A^2 + 2 \left(1 + \frac{1}{\beta}\right) A \gamma + \left(\frac{1}{\beta} - 1\right)^2 \gamma^2},$$

where the last step multiplies numerator and denominator by $\beta\gamma/2$ and uses $(1+1/\beta)^2 - 4/\beta = (1/\beta - 1)^2$. This is $\gamma/(a_1\gamma + a_2 + (a_3\gamma + a_4\gamma^2 + a_5)^{1/2})$ with $a_1 = \frac{1+\beta}{2} = .975$, $a_2 = \frac{\beta A}{2} = .0329$, $a_3 = \frac{\beta(1+\beta)A}{2} = .0642$, $a_4 = (\frac{1-\beta}{2})^2 = .000625$, $a_5 = a_2^2 = .0011$. These are exactly the book's constants, but the book prints γ^{-1} where γ belongs (see the misprint box in the overview). $\lambda_2 \rightarrow 0$ as $\gamma \rightarrow 0$ and $\lambda_2 \rightarrow 1$ as $\gamma \rightarrow \infty$.

(d) (i) The steady state is that of (b): $\hat{k}, \hat{c}, \bar{R} = 1 + \rho$.

(ii) Before $t = 10$ the Euler equation is undistorted except at the date the tax first applies: the return from 9 to 10 is $\bar{R}_{10} = 1 + (1 - \bar{\tau}_k)(f'(k_{10}) - \delta)$. Capital starts to fall at once, because the household spreads the decumulation over time. As k falls, f' rises, so $\bar{R}_{t+1} = f'(k_{t+1}) + 1 - \delta > 1 + \rho$ for $t + 1 \leq 9$ and consumption rises from $t = 0$ to $t = 9$ (after a small jump up at $t = 0$). At $t = 10$ the tax pushes $\bar{R}_{10}$ below $1 + \rho$: consumption falls and keeps declining. k keeps falling toward $\hat{k}$ and $\bar{R}$ rises back to $1 + \rho$ (figure 11.9.5).

(iii) Higher γ : the feedforward discount factor $\lambda_1^{-1} = \beta\lambda_2$ is larger, so distant tax changes matter more today: anticipation starts earlier and is spread more smoothly over $0, \dots, 9$. After $t = 10$ convergence is slower (λ_2 larger). Low γ : little happens until shortly before $t = 10$, then consumption rises sharply and drops at 10; after 10 convergence is fast.

Numerical exercise — see the companion code

$\bar{g} = .2$, $\tau_k = .2$, benchmark otherwise. (a) $\bar{k} = 1.489956$, $\bar{c} = .642645$, $\bar{R} = 1.052632$; (b) $\hat{k} = 1.381220$, $\hat{c} = .636222$.

	$\gamma = .2$	$\gamma = 2$	$\gamma = 5$
λ_2 : root; corrected (11.10.16)	.57788; .57791	.85239; .85243	.91087; .91089
λ_2 as printed in (11.10.16)	.0396	3.534	19.48
(b) $c_0 / k_5 / \bar{R}_1$	.6869 / 1.3894 / 1.0462	.6575 / 1.4334 / 1.0435	.6515 / 1.4525 / 1.0429
(d) $c_0 / c_9 / c_{10}$	.6428 / .6705 / .6566	.6449 / .6502 / .6483	.6451 / .6470 / .6462
(d) $k_{10} / \bar{R}_9 / \bar{R}_{10}$	1.4250 / 1.0567 / 1.0482	1.4423 / 1.0570 / 1.0466	1.4514 / 1.0564 / 1.0457
(d) half-life of $k_t - \hat{k}$ after $t = 10$	2	5	9

Checks: Newton vs shooting (10^{-7}); λ_2 from the numerically differentiated H of (11.6.4) equals the root. The linear approximation (11.10.8) vs Newton: the maximum error relative to the maximum deviation is .5–.7% for the unforeseen tax and 4.7–14.9% for the foreseen one at $\tau_k = .2$. It falls tenfold with each tenfold smaller tax change ($\tau_k = .02, .002$), as a first-order approximation should. The foreseen case is less accurate because the linearisation is around the *terminal* steady state, and the pre-10 policy is far from it.

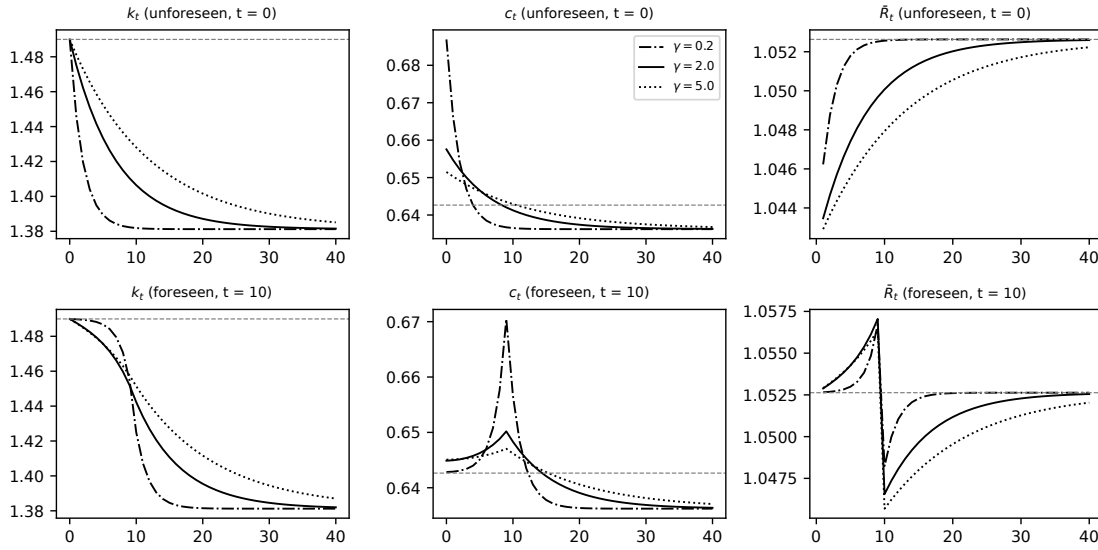


Figure 6: Exercise 11.7: $\tau_k = .2$ unforeseen at $t = 0$ (top) and foreseen for $t = 10$ (bottom); $\gamma = .2$ (dash-dot), 2 (solid), 5 (dotted).

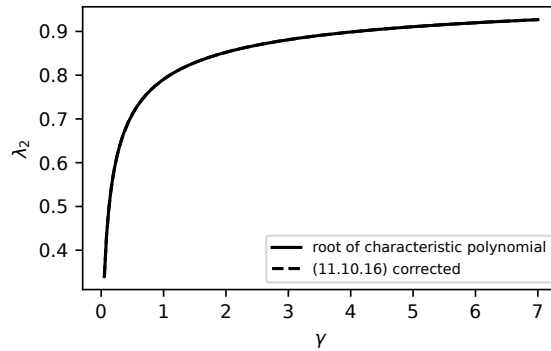


Figure 7: Exercise 11.7(c): $\lambda_2(\gamma)$, root of the characteristic polynomial vs corrected (11.10.16).

Exercise 11.8 Trade and growth, version II

Question

p. 460. **Part I.** Consider the problem of the planner in a small economy. When the economy is closed to trade, the planner chooses $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ to maximize $\sum_{t=0}^{\infty} \beta^t u(c_t)$, $0 < \beta < 1$, $\beta = \frac{1}{1+\rho}$, $\rho > 0$ subject to

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t, \quad \delta \in (0, 1)$$

where

Let $\bar{k}$ be the steady state value of k_t under the optimal plan.

a. Find a formula for $\bar{k}$.

b. Assume that $k_0 > \bar{k}$. Describe time paths for $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ and $\bar{R}_{t+1} = (1 - \delta) + f'(k_{t+1})$.

c. What is the steady state value of $\bar{R}_{t+1}$?

d. Is $\bar{R}_{t+1}$ less or greater than its steady state value when $k_{t+1} > \bar{k}$?

Part II. e. Now assume that the economy is open to international trade in capital and financial assets. Assume that there is a fixed world gross rate of return $R = \beta^{-1}$ at which the planner can borrow or lend. In particular, the planner is free to use the following plan. The planner can sell all of its capital $\bar{k}_0$ and simply consume the interest payments. Let $\bar{k}_0$ be the level of initial capital ($\bar{k}_0 > \bar{k}$) just before the country opens up to trade just before time 0. Let $\bar{k}_0$ be the initial capital per capita $\bar{k}_0 > \bar{k}$ studied in parts a-d. At time $t = -1$, after $\bar{k}_0$ was set, trade opens up. At time $t = -1$, the planner sells $\bar{k}_0$ in exchange for IOU's or bonds from the rest of the world in the amount $A_{-1} = \bar{k}_0$.

The bonds A_{-1} are one-period in duration and bear the constant world gross interest rate $R = \beta^{-1}$. After the sale of $\bar{k}_0 = A_{-1}$, the planner has zero capital and so shuts down the technology. Instead, the planner uses the asset market to smooth consumption. The planner chooses $\{A_{t+1}, c_t\}$ to maximize $\sum_{t=0}^{\infty} \beta^t u(c_t)$, $0 < \beta < 1$, $\beta = \frac{1}{1+\rho}$, $\rho > 0$ subject to $c_t + \beta A_{t+1} = A_t$, $A_0 = \bar{k}_0$. Please find the optimal path for consumption $\{c_t\}$.

f. Compare the path of $(c_t, k_{t+1}, \bar{R}_t)$ that you computed in parts a-d with “no trade” with the part e path “with trade” in which the government “shuts down the home technology” and lives entirely from the returns on foreign assets. Can you say which path the representative consumer will prefer?

g. Now return to the economy in part e with $\bar{k}_0 > \bar{k}$ from part d. Assume that the planner is free to borrow or lend capital at the fixed gross interest of $\beta^{-1} = 1 + \rho$, as before. But now assume the planner chooses the optimal amount of $\bar{k}_0$ to sell off and so does not necessarily sell off the entire $\bar{k}_0$ and possibly continues to operate the technology.

i) Find the solution of the planning problem.

ii) Explain why it is optimal not to shut down the technology. **Hint:** Starting from having shut the technology down, think of putting a small amount ϵ of capital into the technology—this earns $z\epsilon^\alpha$ and costs $\rho\epsilon$ in terms of foregone interest. Because $0 < \alpha < 1$, $\rho\epsilon < z\epsilon^\alpha$ for small ϵ . Thus, the technology is very productive for small ϵ , so it is efficient to use it.

Solution: *Misprints/omissions.* The constraint “ $c_t + k_t + 1$ ” should read $c_t + k_{t+1}$. The sentence “where” is left unfinished; we use the u and f of exercise 11.6 ($u = c^{1-\gamma}/(1-\gamma)$, $f = zk^\alpha$).

(a) As in 11.6: $\bar{k} = (\alpha z/(\rho + \delta))^{1/(1-\alpha)}$.

(b) Mirror image of 11.6(b): with $k_0 > \bar{k}$, k_t falls monotonically to $\bar{k}$. Along the path $f'(k_{t+1}) < \rho + \delta$, so $\bar{R}_{t+1} < 1 + \rho$ and $c_{t+1}/c_t = (\beta \bar{R}_{t+1})^{1/\gamma} < 1$: consumption falls monotonically to $\bar{c}$. $\bar{R}_{t+1}$ rises monotonically to $1 + \rho$.

(c) $1 + \rho$. (d) Less, since f' is decreasing.

(e) Lagrangian $\sum_t \beta^t \{u(c_t) + \nu_t [A_t - c_t - \beta A_{t+1}]\}$. FOCs: $u'(c_t) = \nu_t$; A_{t+1} : $-\beta^{t+1} \nu_t + \beta^{t+1} \nu_{t+1} = 0$. So $u'(c_t) = u'(c_{t+1})$: consumption is constant. Iterate the constraint: $A_0 = \sum_{t=0}^T \beta^t c_t + \beta^{T+1} A_{T+1}$; with no Ponzi scheme, $\beta^T A_T \rightarrow 0$ and

$$\sum_t \beta^t c_t = A_0 = \bar{k}_0 \Rightarrow \boxed{c_t = (1 - \beta) \bar{k}_0 = \frac{\rho}{1+\rho} \bar{k}_0, \quad A_t = \bar{k}_0.}$$

Timing. The constraint as printed sets $A_0 = \bar{k}_0$, so the bonds bought at $t = -1$ earn no interest between -1 and 0 . With the timing of 11.6 (bonds bought at $t = -1$ pay R

at $t = 0$), $A_0 = R\bar{k}_0$ and $c = \rho\bar{k}_0$.

(f) The closed path starts high (c_0 above $\bar{c}$, since the excess capital is eaten) and declines to $\bar{c} = f(\bar{k}) - \delta\bar{k}$, with $\bar{R} < 1 + \rho$ rising. The sell-all path gives a flat $c = (1 - \beta)\bar{k}_0$, with $k = 0$. The ranking depends on $\bar{k}_0$. Selling everything gives up production, and in particular all labour income $w = f(k) - kf'(k)$, in exchange for interest on $\bar{k}_0$. A sufficient condition for the closed economy to be better: it can keep $k_t = \bar{k}_0$ and consume $f(\bar{k}_0) - \delta\bar{k}_0$ forever, which beats $(1 - \beta)\bar{k}_0$ when $f(\bar{k}_0)/\bar{k}_0 \geq \delta + 1 - \beta$, i.e. $\bar{k}_0 \leq (z/(\delta + 1 - \beta))^{1/(1-\alpha)}$. For very large $\bar{k}_0$ the net return $f'(k) - \delta$ is negative, and selling out can win.

(g) (i) With 11.6's timing: sell $\bar{k}_0 - k_0$ at $t = -1$, so $B_{-1} = k_0 - \bar{k}_0$ (negative: the country lends). The FOCs of 11.6 Part II apply unchanged: $k_t = \bar{k}$ for all $t \geq 0$, $B_t = B_{-1} = \bar{k} - \bar{k}_0 < 0$ (constant foreign assets), and

$$c = f(\bar{k}) - \delta\bar{k} + \rho(\bar{k}_0 - \bar{k}) \quad \text{for all } t.$$

(ii) With capital k kept constant and $\bar{k}_0 - k$ lent abroad, consumption is $c(k) = f(k) - (\rho + \delta)k + \rho\bar{k}_0$. Shutting down is $k = 0$, $c(0) = \rho\bar{k}_0$. But $c'(k) = f'(k) - (\rho + \delta) \rightarrow +\infty$ as $k \rightarrow 0$: a small ϵ of capital yields $z\epsilon^\alpha$ and costs $(\rho + \delta)\epsilon$ (foregone interest *plus depreciation*; the hint's $\rho\epsilon$ omits $\delta\epsilon$, which does not change the argument), and $z\epsilon^\alpha > (\rho + \delta)\epsilon$ for small ϵ because $\alpha < 1$. $c(k)$ is maximised at $f'(k) = \rho + \delta$, i.e. $k = \bar{k}$, and $c(\bar{k}) - c(0) = f(\bar{k}) - (\rho + \delta)\bar{k} = (1 - \alpha)f(\bar{k}) > 0$ because $(\rho + \delta)\bar{k} = f'(\bar{k})\bar{k} = \alpha f(\bar{k})$.

Numerical exercise — see the companion code

$z = 1$, $\bar{k}_0 = 2\bar{k} = 2.979913$. Closed: $c_0 = 1.116898$, $\bar{R}_1 = .969594$ (monotone paths checked). Sell-all: $c = (1 - \beta)\bar{k}_0 = .148996$ (or $\rho\bar{k}_0 = .156838$ with 11.6's timing). Optimal open: $c = .921064$. Welfare: sell-all $-134.23 < \text{closed } -22.175 < \text{optimal open } -21.714$. Selling everything beats autarky only if $\bar{k}_0 > 15.505\bar{k} = 23.10$ (**brentq** on the welfare difference). The sufficient condition above gives $\bar{k}_0 \leq (1/.25)^{1/.67} = 7.93$. $\arg \max_k [f(k) - (\rho + \delta)k] = \bar{k}$ checked on a grid.

Exercise 11.9 Linear utility, finite horizon

Question

p. 461. Consider a consumer who wants to choose $\{c_t\}_{t=0}^T$ to maximize $\sum_{t=0}^T \beta^t c_t$, $0 < \beta < 1$ subject to the intertemporal budget constraint

$$\sum_{t=0}^T R^{-t} [c_t - y_t] = 0, \quad R > 1$$

where $c_t \geq 0$ and $y_t > 0$ for $t = 0, \dots, T$. Here $R > 1$ is the gross interest rate $(1 + r)$, $r > 0$. Assume R is constant. Here $\{y_t\}_{t=0}^T$ is an exogenous sequence of “labor income”.

- Assume that $\beta R < 1$. Find the optimal path $\{c_t\}_{t=0}^T$.
- Assume that $\beta R > 1$. Find the optimal path $\{c_t\}_{t=0}^T$.
- Assume that $\beta R = 1$. Find the optimal path $\{c_t\}_{t=0}^T$.

Solution: Let $W = \sum_{t=0}^T R^{-t} y_t > 0$ and change variables to $d_t = R^{-t} c_t \geq 0$. The problem is

$$\max \sum_{t=0}^T (\beta R)^t d_t \quad \text{s.t.} \quad \sum_{t=0}^T d_t = W, \quad d_t \geq 0,$$

a linear program over a simplex: put all of W on the date with the largest weight $(\beta R)^t$. KKT: with multiplier λ on the budget and $\nu_t \geq 0$ on $c_t \geq 0$, $\beta^t - \lambda R^{-t} + \nu_t = 0$, $\nu_t c_t = 0$. So $\lambda \geq (\beta R)^t$ for all t , with equality where $c_t > 0$: $\lambda = \max_t (\beta R)^t$.

(a) $\beta R < 1$: $(\beta R)^t$ is largest at $t = 0$: $c_0 = \sum_{t=0}^T R^{-t} y_t$, $c_t = 0$ ($t \geq 1$). The consumer borrows against all future income and consumes at once.

(b) $\beta R > 1$: largest at $t = T$: $c_T = R^T W = \sum_{t=0}^T R^{T-t} y_t$, $c_t = 0$ ($t < T$). The consumer saves everything and consumes at the end.

(c) $\beta R = 1$: the objective equals $\sum_t R^{-t} c_t = W$ for *every* feasible path. Any $c \geq 0$ that satisfies the budget is optimal, e.g. $c_t = y_t$, or the flat path $c_t = W(1 - R^{-1})/(1 - R^{-(T+1)})$. With linear utility there is no motive to smooth; a strictly concave u would select the flat path through $u'(c_t) = \beta R u'(c_{t+1})$.

Numerical exercise — see the companion code

$T = 5$, $R = 1.05$, y_t drawn once from $U(.5, 1.5)$ (seed 0), $W = 4.994328$. **linprog:** $\beta = .9$: $c_0 = 4.9943$; $\beta = 1/R + .02$: $c_5 = 6.3742 = R^5 W$; $\beta = 1/R$: optimal value 4.994328 = W , also attained by $c = y$ and by a perturbed feasible path.

Reading Paths and Yield Curves (11.10–11.12)

Exercise 11.10 Term structure of interest rates

Question

p. 462. This problem assumes the following environment. A representative consumer has preferences ordered by $\sum_{t=0}^{\infty} \beta^t u(c_t)$, $0 < \beta < 1$, $\beta = \frac{1}{1+\rho}$, $\rho > 0$, where c_t is the consumption per worker and where $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ if $\gamma > 0$ and $\gamma \neq 1$, $\log(c)$ if $\gamma = 1$. The technology is $y_t = f(k_t) = zk_t^\alpha$, $0 < \alpha < 1$, $z > 0$, where y_t is the output per unit labor and k_t is capital per unit labor, $y_t = c_t + x_t + g_t$, $k_{t+1} = (1 - \delta)k_t + x_t$, $0 < \delta < 1$, where x_t is gross investment per unit of labor and g_t is government expenditures per unit of labor. The government finances its expenditures by levying some combination of a flat rate tax τ_{ct} on the value of consumption goods purchased at t , a flat rate tax τ_{nt} on the value of labor earnings at t , a flat rate tax τ_{kt} on earnings from capital at t and a lump sum tax of τ_{ht} in time t consumption goods per worker at time t . Define $\bar{R}_{t+1} = \frac{(1+\tau_{ct})}{(1+\tau_{ct+1})} [1 + (1 - \tau_{kt+1})(f'(k_{t+1}) - \delta)]$.

a. Recall that we can represent $q_t^0 = q_0^0 m_{0,1} m_{1,2} \cdots m_{t-1,t}$ where $m_{t-1,t} = \frac{q_t^0}{q_{t-1}^0}$ and $m_{t-1,t} \equiv \exp(-r_{t-1,t}) \approx \frac{1}{1+r_{t-1,t}}$. Further, recall that the t period long yield satisfies $q_t^0 = \exp(-tr_{0,t})$ and $r_{0,t} = \frac{1}{t}[r_{0,1} + r_{1,2} + \cdots + r_{t-1,t}]$. Now suppose that at $t = 0$, $k_0 = \bar{k}$, where $\bar{k}$ is the steady state appropriate for an economy with constant $g_t = \bar{g} > 0$ and all expenditures are financed by lump sum taxes. Find q_t^0 for this economy.

b. Plot $r_{t-1,t}$ for this economy for $t = 1, 2, \dots, 10$.

c. Plot $r_{0,t}$ for this economy for $t = 1, 2, \dots, 10$ (this is what Bloomberg plots).

d. Now assume that at time 0, starting from $k_0 = \bar{k}$ for the steady state you computed in part a, the government unexpectedly and permanently raises the tax rate on income from capital $\tau_{kt} = \tau_k > 0$ to a positive rate.

i) Plot $r_{t-1,t}$ for this economy for $t = 1, 2, \dots, 10$. Explain how you got this outcome.

ii) Plot $r_{0,t}$ for this economy for $t = 1, 2, \dots, 10$. Explain how you got this outcome.

Solution: *Misprint.* In the definition of $\bar{R}_{t+1}$, “ $f'(k_{t+1}) - \delta$ ” should be $f'(k_{t+1}) - \delta$, as in (11.6.8e). We use $r_{t-1,t} \equiv -\log m_{t-1,t}$ exactly.

(a) With lump-sum finance and $k_0 = \bar{k}$, the equilibrium is stationary (as in 11.5(c)): $c_t = \bar{c}$, $k_t = \bar{k}$. By (11.6.8b), $q_t^0 = \beta^t u'(\bar{c}) / (1 + \tau_c)$; normalising $q_0^0 = 1$,

$$q_t^0 = \beta^t, \quad m_{t-1,t} = \beta.$$

(b) $r_{t-1,t} = -\log \beta = \log(1 + \rho) \approx \rho$ for every t : a flat line.

(c) $r_{0,t} = t^{-1} \sum_{s=1}^t r_{s-1,s} = -\log \beta$: the yield curve is flat.

(d) After the surprise, the economy follows the path of 11.7(b): k declines from $\bar{k}$ to $\hat{k}$, c jumps up and declines, and $\bar{R}_{t+1} < 1 + \rho$ rises back to $1 + \rho$.

(i) By (11.6.8f) with τ_c constant, $m_{t,t+1} = \beta u'(c_{t+1}) / u'(c_t) = 1 / \bar{R}_{t+1}$, so

$$r_{t,t+1} = \log \bar{R}_{t+1} = \log \left[1 + (1 - \tau_k)(f'(k_{t+1}) - \delta) \right] = -\log \beta + \gamma \log \frac{c_{t+1}}{c_t}.$$

Since c falls along the transition, short rates are below $-\log \beta$. Since the fall slows down as $k \rightarrow \hat{k}$, they rise monotonically toward $-\log \beta$. Equivalently: k_{t+1} falls toward $\hat{k}$, so $f'(k_{t+1})$ rises toward $\delta + \rho/(1 - \tau_k)$.

(ii) $r_{0,t}$ is the running average of the short rates. The short rates increase, so the yield curve slopes upward, lies below the short-rate curve for $t \geq 2$, and converges to $-\log \beta$ at long maturities.

Numerical exercise — see the companion code

Benchmark, $\bar{g} = .2$, $\tau_k = .2$. (a)–(c) $q_t = \beta^t$, $r = -\log \beta = .051293$ (checked on the Newton path). (d) Check: $\log(q_t/q_{t+1}) = \log \bar{R}_{t+1}$ on the Newton path.

t	1	2	3	4	5	6	7	8	9	10
$r_{t-1,t}$	.04255	.04370	.04469	.04556	.04632	.04698	.04755	.04805	.04848	.04886
$r_{0,t}$	.04255	.04312	.04365	.04413	.04457	.04497	.04534	.04568	.04599	.04627

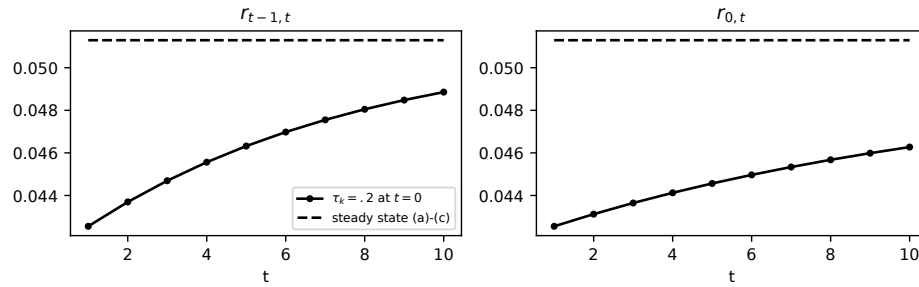


Figure 8: Exercise 11.10: short rates and yields; flat at $-\log \beta$ in (a)–(c), rising after the capital tax in (d).

Exercise 11.11 A rising consumption path

Question

p. 463. This problem assumes the same economic environment as the previous exercise i.e., the “growth model” with fiscal policy. Suppose that you observe the path for consumption per capita in figure 11.1. Say what you can about the likely behavior over time of k_t , $\bar{R}_t = [1 + (1 - \tau_{kt})(f'(k_t) - \delta)]$, g_t and τ_{kt} . (You are free to make up any story that is consistent with the model.)

[Figure 11.1: consumption per capita c_t starts at a positive level at $t = 0$ and rises at a decreasing rate.]

Solution: Keep τ_c constant. Then (11.6.9) gives

$$\log \frac{c_{t+1}}{c_t} = \frac{1}{\gamma} [\log \beta + \log \bar{R}_{t+1}].$$

In figure 11.1 consumption growth is positive and declining, so $\bar{R}_{t+1} > 1 + \rho$ and $\bar{R}_{t+1}$

falls toward $1 + \rho$. With $\bar{R}_t = 1 + (1 - \tau_{kt})(f'(k_t) - \delta)$ and a constant τ_k , a falling $\bar{R}$ means a falling $f'(k_t)$, i.e. *rising capital*, converging to its steady state. Two consistent stories:

- (i) *No policy change*: constant g_t and τ_{kt} , and the economy starts with k_0 below its steady state (the standard transition of 11.6(b) and 11.14(c)).
- (ii) *A surprise capital-tax cut*: the economy was in the steady state with $\tau_k > 0$. At $t = 0$ the government unexpectedly and permanently sets $\tau_{kt} = 0$ (constant g_t , lump-sum taxes absorb the revenue loss). The after-tax return at the old capital stock jumps above $1 + \rho$; the household invests; k rises to the higher steady state $f'(k) = \rho + \delta$; $\bar{R}$ falls back to $1 + \rho$.

In both, g_t is constant; τ_{kt} is constant from $t = 0$ on (in (ii), at a lower level than before $t = 0$). What is *not* consistent: a surprise permanent change in g alone (consumption shifts once and stays flat, 11.5(c)); an anticipated future rise in g (consumption falls, 11.17).

Numerical exercise — see the companion code

Checked on the Newton paths (first 40 periods): $\Delta c_t > 0$ and decreasing; $\bar{R}_t > 1 + \rho$ and decreasing. (i) $k_0 = .5\bar{k}$: $c_0 = .4665 \rightarrow .6426$, $k_0 = .7450 \rightarrow 1.4900$, $\bar{R}_1 = 1.1718$. (ii) $\tau_k = .2 \rightarrow 0$ at $t = 0$: $c_0 = .6204 \rightarrow .6426$, $k_0 = 1.3812 \rightarrow 1.4900$, $\bar{R}_1 = 1.0638$.

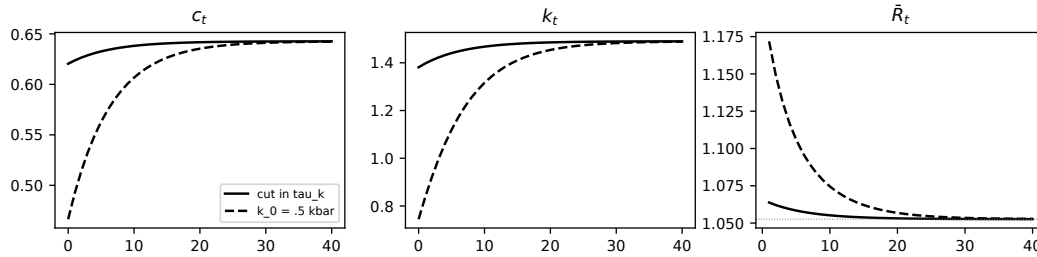


Figure 9: Exercise 11.11: surprise cut in τ_k (solid) and $k_0 = .5\bar{k}$ (dashed).

Exercise 11.12 Flat, then a drop, then a decline

Question

p. 463. Assume the same economic environment as in the previous two problems. Assume that someone has observed the time path for c_t in figure 11.2:

[Figure 11.2: c_t is constant from $t = 0$ until $t = 12$, drops discretely at $t = 12$, and then declines at a decreasing rate toward a lower level.]

a. Describe a consistent set of assumptions about the fiscal policy that explains this time path for c_t . In doing so, please distinguish carefully between changes in taxes and expenditures that are foreseen versus unforeseen.

b. Describe what is happening to k_t , $\bar{R}_t$ and g_t over time. Make whatever assumptions you must to get a complete but consistent story - here “consistent” means “consistent with the economic environment we have assumed”.

Solution:

(a) Before $t = 12$. Constant consumption means $\bar{R}_{t+1} = 1 + \rho$ by (11.6.9): the economy sits in a steady state. Nothing that happens at or after $t = 12$ was known before $t = 12$: a foreseen change would move consumption before it occurs (section 11.9). So the news arrives at $t = 12$ and is *unforeseen*.

At $t = 12$. A discrete drop is a negative wealth effect: the present value of taxes jumps (higher government spending). After $t = 12$. Gradual decline means $\bar{R}_{t+1} < 1 + \rho$ for a while, rising back to $1 + \rho$. Two consistent stories:

- **A (unforeseen news of a foreseen change).** At $t = 12$ the government announces, unexpectedly, a permanent rise in g from $t = 22$ on, financed by lump-sum taxes. From $t = 12$ on the rise is foreseen. Consumption drops at 12 (wealth effect) and then declines smoothly to $\bar{c} - \Delta g$. This is figure 11.9.1 shifted to start at $t = 12$.
- **B (unforeseen immediate change with a distorting tax).** At $t = 12$, g rises permanently and at once and the government raises τ_k permanently (lump sums cover the rest). Consumption drops at 12, by a little less than Δg because the tax increase also makes the household run down capital. It then declines as k falls toward the lower steady state of the higher τ_k .

(b) Story A: $g = .2$ until $t = 21$, $.4$ from $t = 22$; $\tau_k = 0$. After 12, capital is *built up* until $t = 22$ and then run down back to $\bar{k}$ (the steady state does not depend on g). $\bar{R}$ falls below $1 + \rho$ after 12, is lowest at $t = 22$ when k peaks, and then rises back to $1 + \rho$. Story B: g jumps at 12; k declines monotonically from $\bar{k}$ to $\hat{k}$; $\bar{R}$ drops below $1 + \rho$ at 13 and rises back. The two stories differ in the path of k (up then down in A; down in B), which data on investment could separate.

Numerical exercise — see the companion code

Benchmark; A: $g = .2 \rightarrow .4$ at $t = 22$, announced at 12; B: $g = .2 \rightarrow .4$ and $\tau_k = 0 \rightarrow .2$ at 12. Both reproduce the shape (checked: $c_{12} < c_{11}$ and c strictly decreasing from 12 to 51). A: $c_{11} = .6426$, $c_{12} = .6092$, $c_{40} = .4518$; $\max_t k_t = 2.0985$ (at $t = 22$); $\bar{R}_{13} = 1.0489$. B: $c_{12} = .4548$, $c_{40} = .4370$; $k_{40} = 1.3854$; $\bar{R}_{13} = 1.0432$.

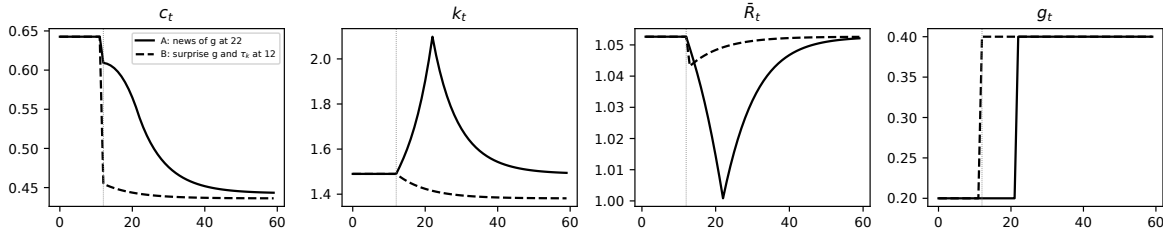


Figure 10: Exercise 11.12: story A (solid) and story B (dashed); vertical line at $t = 12$.

Linear Utility, a Transition with g , Yield Curves and News (11.13–11.17)

Exercise 11.13 Optimal growth with linear utility

Question

p. 464. Consider the optimal growth model with a representative consumer with preferences $\sum_{t=0}^{\infty} \beta^t c_t$, $0 < \beta < 1$, with technology

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t, \quad \delta \in (0, 1), \quad c_t \geq 0, \quad k_0 > 0 \text{ given}$$

$$f' > 0, \quad f'' < 0, \quad \lim_{k \rightarrow 0} f'(k) = +\infty, \quad \lim_{k \rightarrow +\infty} f'(k) = 0.$$

Let $\bar{k}$ be the steady state capital stock for the optimal planning problem.

- For k_0 given, formulate and solve the optimal planning problem.
- For $k_0 > \bar{k}$, describe the optimal time path of $\{c_t, k_{t+1}\}_{t=0}^{\infty}$.
- For $k_0 < \bar{k}$, describe the optimal path of $\{c_t, k_{t+1}\}_{t=0}^{\infty}$.
- Let the saving rate s_t be defined as the s_t that satisfies $k_{t+1} = s_t f(k_t) + (1 - \delta)k_t$. Here s_t in general varies along a $\{c_t, k_{t+1}\}$ sequence. Say what you can about how s_t varies as a function of k_t .

Solution: Let $Y(k) = f(k) + (1 - \delta)k$ (increasing). Define $\bar{k}$ by $f'(\bar{k}) = \rho + \delta$.

(a) $\max \sum_t \beta^t [Y(k_t) - k_{t+1}]$ subject to $0 \leq k_{t+1} \leq Y(k_t)$ (the upper bound is $c_t \geq 0$).

Regroup the terms by k_t :

$$\sum_{t \geq 0} \beta^t [Y(k_t) - k_{t+1}] = Y(k_0) + \sum_{t \geq 1} \beta^t \varphi(k_t), \quad \varphi(k) \equiv Y(k) - \frac{k}{\beta}.$$

$\varphi'(k) = f'(k) + 1 - \delta - 1/\beta = f'(k) - (\rho + \delta)$, so φ is strictly concave, increasing on $(0, \bar{k})$ and maximised at $\bar{k}$. The objective is separable in the k_t ; only the constraints $k_{t+1} \leq Y(k_t)$ link dates. Claim:

$$k_{t+1} = \min\{Y(k_t), \bar{k}\}, \quad c_t = \max\{0, Y(k_t) - \bar{k}\}.$$

Proof. Let $\{k_t^*\}$ be this path and $\{k_t\}$ any feasible path, both from k_0 . (1) If $k_t^* < \bar{k}$ then $k_t \leq k_t^*$. By induction: if $k_t^* < \bar{k}$, then $k_t^* = Y(k_{t-1}^*)$ with $k_{t-1}^* < \bar{k}$, and $k_t \leq Y(k_{t-1}) \leq Y(k_{t-1}^*) = k_t^*$ since $k_{t-1} \leq k_{t-1}^*$ and Y is increasing. (2) Hence at dates with $k_t^* < \bar{k}$, $\varphi(k_t) \leq \varphi(k_t^*)$ because φ is increasing below $\bar{k}$. At dates with $k_t^* = \bar{k}$, $\varphi(k_t^*) = \max \varphi \geq \varphi(k_t)$. Summing, $\{k_t^*\}$ is optimal (uniquely, by strict concavity of φ on the dates where it binds). Value for $Y(k) \geq \bar{k}$: $V(k) = Y(k) - \bar{k} + \beta \bar{c}/(1 - \beta)$ with $\bar{c} = f(\bar{k}) - \delta \bar{k}$.

The Euler equation explains the solution: with linear utility, $c_t > 0$ requires $\beta[f'(k_{t+1}) + 1 - \delta] = 1$, i.e. $k_{t+1} = \bar{k}$. If the return exceeds $1/\beta$ ($k_{t+1} < \bar{k}$) the planner saves everything; if it falls short, it consumes the excess.

(b) $k_0 > \bar{k}$: $Y(k_0) > \bar{k}$, so $c_0 = Y(k_0) - \bar{k}$ (the excess capital is eaten at once); then $k_t = \bar{k}$ and $c_t = \bar{c}$ for $t \geq 1$.

(c) $k_0 < \bar{k}$: while $Y(k_t) < \bar{k}$, $c_t = 0$ and $k_{t+1} = Y(k_t)$ (grow as fast as possible). At the first date τ with $Y(k_\tau) \geq \bar{k}$: $c_\tau = Y(k_\tau) - \bar{k} \geq 0$ and $k_{\tau+1} = \bar{k}$; then $\bar{c}$ forever. This is a “most rapid approach” path: with linear utility the elasticity of intertemporal substitution is infinite ($\gamma = 0$, so $\lambda_2 = 0$ in (11.10.16)).

(d) $s_t = [k_{t+1} - (1 - \delta)k_t]/f(k_t)$. With the policy of (a), let $\underline{k}$ solve $Y(\underline{k}) = \bar{k}$.

$$s(k) = \begin{cases} 1, & k \leq \underline{k}, \\ \frac{\bar{k} - (1 - \delta)k}{f(k)}, & k \geq \underline{k}. \end{cases}$$

s is continuous at $\underline{k}$ ($s = 1$). On $[\underline{k}, \infty)$, $s'(k) = \{-(1 - \delta)f - [\bar{k} - (1 - \delta)k]f'\}/f^2$. Where the numerator of s is positive, both terms are negative. Where it is negative, $-(1 - \delta)f + ((1 - \delta)k - \bar{k})f' = -(1 - \delta)(f - kf') - \bar{k}f' < 0$, since $f - kf' > 0$. So s equals 1 for low k and then decreases strictly. At the steady state $s(\bar{k}) = \delta \bar{k}/f(\bar{k}) (= \delta \alpha/(\rho + \delta))$ for Cobb–Douglas. $s < 0$ (dissaving, consuming out of capital) for $k > \bar{k}/(1 - \delta)$.

Numerical exercise — see the companion code

Benchmark $f = k^{.33}$: $\bar{k} = 1.489956$, $\bar{c} = .842645$, $\underline{k} = .733812$, $s(\bar{k}) = .261250$, $s < 0$ for $k > 1.862446$. Grid VFI (1501 points including $\bar{k}$; 534 iterations) confirms: policy $= \bar{k}$ whenever $Y(k) \geq \bar{k}$, $c = 0$ (up to grid spacing) otherwise, and $V(k) = Y(k) - \bar{k} + \beta \bar{c}/(1 - \beta)$ on the reachable region. From $k_0 = .05$: $k = .05, .4121, 1.0760, 1.49, \dots$; $c = 0, 0, .3954, .8426, \dots$

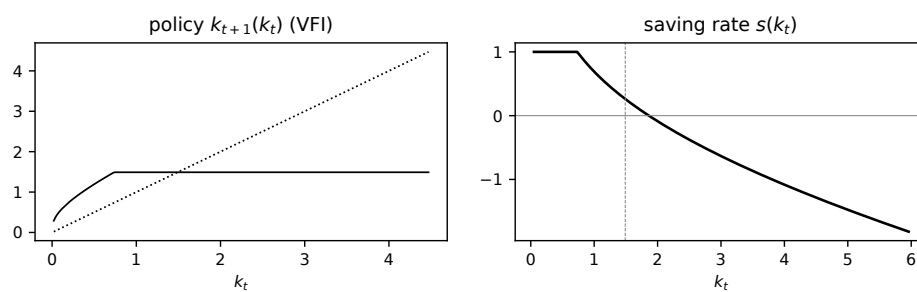


Figure 11: Exercise 11.13: VFI policy function and the saving rate $s(k)$ (dashed: $\bar{k}$).

Exercise 11.14 Transitions with and without government purchases

Question

p. 465. A representative consumer has preferences over consumption streams ordered by $\sum_{t=0}^{\infty} \beta^t u(c_t)$, $0 < \beta < 1$, where $\beta \equiv \frac{1}{1+\rho}$, c_t is consumption per worker, $\gamma > 0$, and $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ if $\gamma > 0$ and $\gamma \neq 1$, $\log(c)$ if $\gamma = 1$. The consumer supplies one unit of labor inelastically. The technology is $y_t = f(k_t) = zk_t^\alpha$, $0 < \alpha < 1$ where y_t is output per worker, k_t is capital per worker, x_t is gross investment per worker, g_t = government expenditures per worker, and $y_t = c_t + x_t + g_t$, $k_{t+1} = (1 - \delta)k_t + x_t$, $0 < \delta < 1$.

The government finances its expenditure stream $\{g_t\}$ by levying a stream of flat rate taxes $\{\tau_{ct}\}$ on the value of the consumption good purchased at t , a stream of flat rate taxes $\{\tau_{kt}\}$ on earnings from capital at t , and a stream of lump sum taxes $\{\tau_{ht}\}$. Let $\{q_t, q_t\eta_t, q_t w_t\}_{t=0}^{\infty}$ be a *price system*, where q_t is the price of time t consumption and investment goods, $q_t\eta_t$ is the price of renting capital at time t , and $q_t w_t$ is the price of renting labor at time t . All trades occur at time 0 and all prices are measured in units of the time 0 consumption good. The initial capital stock k_0 is given.

- a. Define a competitive equilibrium with taxes and government purchases.
- b. Assume that $g_t = 0$ for all $t \geq 0$ and that all taxes are also zero. Find the value $\bar{k}$ of the steady state capital per worker. Find a formula for the saving rate $\frac{x_t}{f(k_t)}$ at the steady state value of the capital stock.
- c. Suppose that the initial capital stock $k_0 = .5\bar{k}$, so that the economy starts below its steady state level. Describe (i.e., draw graphs showing) the time paths of $\{c_t, k_{t+1}, \bar{R}_{t+1}\}_{t=0}^{\infty}$ where $\bar{R}_{t+1} \equiv [(1 - \delta) + f'(k_{t+1})]$.
- d. Starting from the same initial k_0 as in part c, assume now that $g_t = \bar{g} = \phi f(k_0) > 0$ for all $t \geq 0$ where $\phi \in (0, 1 - \delta)$. Assume that the government finances its purchases by imposing lump sum taxes. Describe (i.e., draw graphs) showing the time paths of $\{c_t, k_{t+1}, \bar{R}_{t+1}\}_{t=0}^{\infty}$ and compare them to the outcomes that you obtained in part c. What outcomes differ? What outcomes, if any, are identical across the two economies? Please explain.
- e. Starting from the same initial k_0 assumed in part c, assume now that $g_t = \bar{g} = \phi f(k_0) > 0$ for all $t \geq 0$ where $\phi \in (0, 1 - \delta)$. Assume that the government must now finance these purchases by imposing a time-invariant tax rate $\bar{\tau}_k$ on capital each period. The government *cannot* impose lump sum taxes or any other kind of taxes to balance its budget. Please describe how to find a competitive equilibrium.

Solution:

(a) A competitive equilibrium is a price system $\{q_t, \eta_t, w_t\}$, a policy $\{g_t, \tau_{ct}, \tau_{kt}, \tau_{ht}\}$ with $\sum_t q_t g_t = \sum_t q_t [\tau_{ct} c_t + \tau_{kt} (\eta_t - \delta) k_t + \tau_{ht}]$, and an allocation $\{c_t, x_t, k_{t+1}\}$ such that: the household maximises $\sum_t \beta^t u(c_t)$ subject to $\sum_t q_t [(1 + \tau_{ct}) c_t + x_t] \leq \sum_t q_t [\eta_t k_t - \tau_{kt} (\eta_t - \delta) k_t + w_t - \tau_{ht}]$ and $k_{t+1} = (1 - \delta) k_t + x_t$, k_0 given; the firm sets $\eta_t = f'(k_t)$, $w_t = f(k_t) - k_t f'(k_t)$; and $c_t + x_t + g_t = f(k_t)$. The equilibrium conditions are (11.6.3), (11.6.1), k_0 and (11.5.3).

(b) $f'(\bar{k}) = \rho + \delta$: $\bar{k} = (\alpha z / (\rho + \delta))^{1/(1-\alpha)}$. At the steady state $x = \delta \bar{k}$ and $\alpha z \bar{k}^{\alpha-1} = \rho + \delta$ gives $\bar{k}/f(\bar{k}) = \alpha/(\rho + \delta)$, so

$$\frac{x}{f(\bar{k})} = \frac{\delta \bar{k}}{f(\bar{k})} = \frac{\alpha \delta}{\rho + \delta}.$$

(c) As in 11.6(b): k_t and c_t rise monotonically to $\bar{k}$ and $\bar{c} = f(\bar{k}) - \delta \bar{k}$; $\bar{R}_{t+1} > 1 + \rho$ and falls monotonically to $1 + \rho$ (figure).

(d) Lump-sum finance leaves the Euler equation unchanged and shifts feasibility by $\bar{g}$: $u'(c_t) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]$, $c_t = f(k_t) + (1 - \delta)k_t - \bar{g} - k_{t+1}$.

- *Identical*: the steady-state capital $\bar{k}$ (the Euler equation does not contain g), hence the steady-state $\bar{R} = 1 + \rho$, $\eta = f'(\bar{k})$, w and saving rate; the mapping (11.6.9) from $\bar{R}$ to consumption growth; and the irrelevance of the timing of lump-sum taxes.
- *Different*: the level of consumption (lower by exactly $\bar{g}$ in the steady state; by less than $\bar{g}$ at first, because investment also falls), and the transition. With $g > 0$ capital accumulates *more slowly* and $\bar{R}$ stays higher for longer. Reason: the feedback coefficient depends on $A/\gamma = \beta c|f''|/\gamma$ (see 11.7(c)); lower consumption means lower A and a λ_2 closer to 1. Intuitively, $-u''/u' = \gamma/c$: when consumption is low, a given cut in consumption is a larger proportional cut and more costly in utility, so the household is less willing to sacrifice consumption to build capital.

(e) With no lump-sum taxes, the budget $\sum_t q_t \bar{\tau}_k (f'(k_t) - \delta)k_t = \sum_t q_t \bar{g}$ must hold at the equilibrium prices. Algorithm:

1. Guess $\bar{\tau}_k$. Terminal steady state: $f'(\hat{k}) = \delta + \rho/(1 - \bar{\tau}_k)$.
2. Shoot (or Newton) on (11.6.3) with $\tau_{kt} = \bar{\tau}_k$, $g_t = \bar{g}$, from k_0 to $\hat{k}$: this gives $\{c_t, k_{t+1}\}$.
3. Prices $q_t = \beta^t u'(c_t)/(1 + \tau_c)$, $\eta_t = f'(k_t)$ (normalise $q_0 = 1$).
4. Compute the gap $G(\bar{\tau}_k) = \sum_t q_t \bar{\tau}_k (f'(k_t) - \delta)k_t - \bar{g} \sum_t q_t$ (the $t = 0$ term taxes the given k_0 and does not distort).
5. Update $\bar{\tau}_k$ by bisection until $G = 0$, taking the smallest root (revenue has a Laffer curve); if $\max G < 0$, no equilibrium exists.

Numerical exercise — see the companion code

Benchmark with $g = 0$ in (c); $\phi = .1$, so $\bar{g} = \phi f(k_0) = .090742$ in (d)–(e). ϕ is chosen so that (e) has an equilibrium: the steady-state capital-tax revenue cannot exceed .146 (see 11.2), and the book's range $\phi \in (0, 1 - \delta)$ allows values that cannot be financed. (b) $\bar{k} = 1.489956$, $x/f = .261250$. (c) $k_0 = .744978$: $c_0 = .647540$, $\bar{R}_1 = 1.166272$. (d) $c_0 = .565017$, $\bar{R}_1 = 1.168648$; $k_{10} = 1.342656$ vs 1.360230 without g (k_t lower and $\bar{R}_{t+1}$ higher at every date up to 40, checked); same $\bar{k}$ and limiting $\bar{R}$. (e) $\bar{\tau}_k = .641199$, $\hat{k} = .929016$, $c_0 = .649316$, $k_{10} = .866239$.

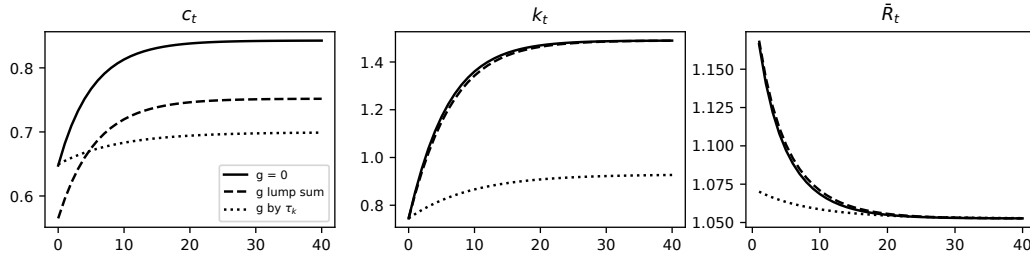


Figure 12: Exercise 11.14: $g = 0$ (solid), g by lump sums (dashed), g by a constant $\bar{\tau}_k$ (dotted).

Exercise 11.15 Reading a yield curve

Question

p. 466. The structure of the economy is identical to that described in the previous exercise. Let $r_{0,t}$ be the yield to maturity on a t period bond at time 0, $t = 1, 2, \dots$. At time 0, Bloomberg reports the term structure of interest rates in figure 11.3. Please say what you can about the evolution of $\{c_t, k_{t+1}\}$ in this economy. Feel free to make any assumptions you need about fiscal policy $\{g_t, \tau_{kt}, \tau_{ct}, \tau_{ht}\}_{t=0}^{\infty}$ to make your answer coherent.

[Figure 11.3: the yield to maturity $r_{0,t}$ as a function of $t = 0, \dots, 40$ starts low and rises at a decreasing rate.]

Solution: From (11.6.8f),

$$r_{t,t+1} = -\log \beta + \gamma \log \frac{c_{t+1}}{c_t} + \log \frac{1 + \tau_{c,t+1}}{1 + \tau_{ct}}, \quad r_{0,t} = \frac{1}{t} \sum_{s=0}^{t-1} r_{s,s+1}.$$

A rising, concave yield curve means that short rates rise over time and settle down. Once policy is constant the long rate must converge to $-\log \beta$, so the short rates are below $-\log \beta$ and rising. With τ_c constant this means $c_{t+1} < c_t$: *consumption falls over time at a decreasing rate and converges*. Because $r_{t,t+1} = \log \bar{R}_{t+1} = \log[1 + (1 - \tau_k)(f'(k_{t+1}) - \delta)] < \log(1 + \rho)$, $f'(k_{t+1})$ is below its steady-state value: *capital is above its steady state and falling*. Consistent stories:

- (i) a surprise permanent rise in τ_k at $t = 0$, starting from the old steady state (11.10(d)): c jumps up at 0 and then declines, k declines to the new, lower steady state;
- (ii) constant policy and k_0 above its steady state (e.g. capital inherited from an earlier regime with higher saving): c and k decline monotonically.

A path of *rising* consumption taxes would also raise short rates (the last term); we do not need it.

Numerical exercise — see the companion code

Checked on the Newton paths: $r_{0,t}$ increasing and concave for $t \leq 40$, c and k decreasing. (i) $\tau_k = .2$ at 0: $r_{0,1} = .04255$, $r_{0,10} = .04627$, $r_{0,40} = .04965 \rightarrow .05129$; $c_0 = .6575 \rightarrow .6362$. (ii) $k_0 = 1.5\bar{k}$: $r_{0,1} = -.00024$, $r_{0,10} = .02171$, $r_{0,40} = .04185$; $c_0 = .7765 \rightarrow .6426$. Story (ii) produces the steeper curve of figure 11.3.

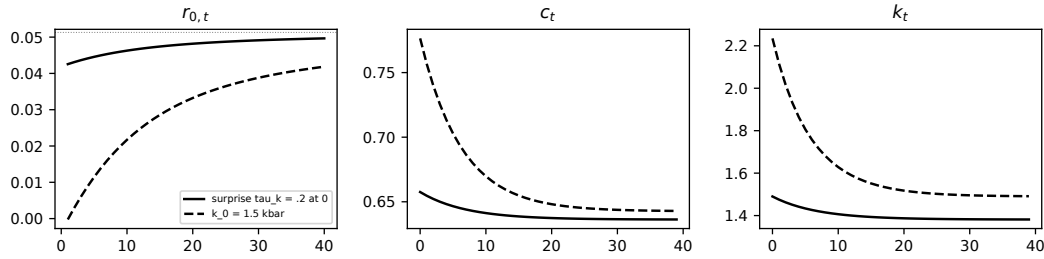


Figure 13: Exercise 11.15: yield curves at $t = 0$ and the implied paths; surprise τ_k (solid), $k_0 = 1.5\bar{k}$ (dashed).

Exercise 11.16 High and low γ

Question

p. 467. The structure of the economy is identical to that described in exercise 11.14. Assume that $\{g_t, \tau_{ct}, \tau_{kt}\}_{t=0}^{\infty}$ are all constant sequences (their values don't change over time). In this problem, we ask you to infer differences across two economies in which all aspects of the economy are identical *except* the parameter γ in the utility function. (It is possible that lump sum taxes differ across the two economies. Assume that lump sum taxes are adjusted to balance the government budget.) In both economies, $\gamma > 0$. In one economy, $\gamma > 0$ is high and in the other it is low. Among other identical features, the two economies have identical government policies and identical initial capital stocks.

a. Please look at figure 11.4. Please tell which outcome for $\{k_{t+1}\}_{t=0}^{\infty}$ describes the low γ economy, and which describes the high γ economy. Please explain your reasoning.

b. Please look at figure 11.5. Please tell which outcome for $\{f'(k_t)\}_{t=0}^{\infty}$ describes the low γ economy, and which describes the high γ economy. Please explain your reasoning.

c. Please plot time paths of consumption for the low γ and the high γ economies.

[Figure 11.4: capital rises from the same k_0 to the same steady state, slowly in panel (a) and very fast in panel (b). Figure 11.5: $f'(k_t)$ falls to $\rho + \delta$, slowly in panel (a) and very fast in panel (b).]

Solution: Constant policies give the same steady state $\bar{k}$ for both economies ((11.6.7)

does not contain γ), the same $\bar{R} = 1 + \rho$ and the same $\bar{c}$. They differ only in the speed of the transition. Linearised, $k_{t+1} - \bar{k} = \lambda_2(k_t - \bar{k})$, and λ_2 increases with γ (11.7(c), corrected (11.10.16)). Equivalently, by (11.6.9) a given gap $\bar{R}_{t+1} - (1 + \rho)$ produces consumption growth $\gamma^{-1} \log(\beta \bar{R}_{t+1})$, large when γ is small.

(a) Panel (a), slow convergence, is the *high- γ* economy. Panel (b), fast convergence, is the *low- γ* economy. With low γ the household does not mind a steep consumption profile: it sacrifices a lot of consumption early and invests heavily, so k reaches $\bar{k}$ quickly. With high γ it smooths consumption and accumulates slowly.

(b) f' is decreasing, so $f'(k_t)$ mirrors k_t : panel (a), slow decline to $\rho + \delta$, is high γ ; panel (b), fast decline, is low γ .

(c) Both consumption paths rise to the same $\bar{c}$. Low γ : c_0 very low (heavy investment), then a steep rise, close to $\bar{c}$ within a few periods. High γ : c_0 higher, then a slow rise. The paths cross: after a few periods the low- γ economy has more capital and consumes more.

Numerical exercise — see the companion code

Benchmark with $g = .2$, $k_0 = .5\bar{k}$; $\gamma = .2$ vs $\gamma = 5$ ($\lambda_2 = .5779$ vs $.9109$). Low γ : $k_5 = 1.4287$, $c_0 = .2945$; high γ : $k_5 = .9780$, $c_0 = .5071$; consumption paths cross at $t = 3$. Newton vs shooting to 10^{-7} .

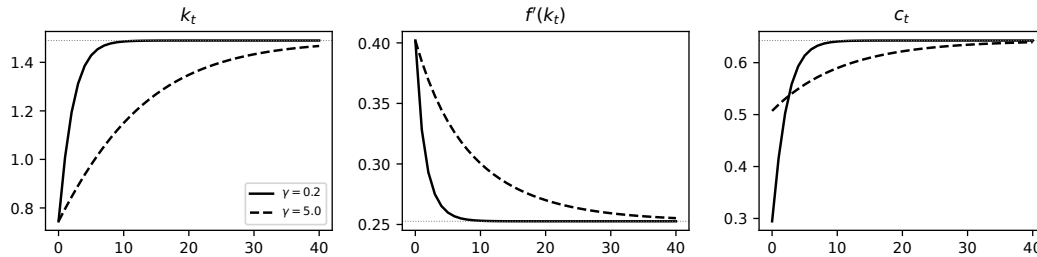


Figure 14: Exercise 11.16: $\gamma = .2$ (solid) vs $\gamma = 5$ (dashed); dotted lines: common steady state.

Exercise 11.17 Anticipated vs unanticipated jump in g

Question

p. 468. The structure of the economy is identical to that described in exercise 11.16. Assume that $\{\tau_{ct}, \tau_{kt}\}_{t=0}^{\infty}$ are constant sequences (their values don't change over time) but that $\{g_t\}_{t=0}^{\infty}$ follows the path described in panel **c** of figure 11.6 – it takes a once and for all jump at time $t = 10$. Lump sum taxes adjust to balance the government budget. Panels **a** and **b** give consumption paths for two economies that are identical except in one respect. In one of the economies, the time 10 jump in g had been anticipated since time 0, while in the other, the jump in g that occurs at time 10 is completely unanticipated at time 10. Please tell which panel corresponds to which view of the arrival of news about the path of g_t . Please say as much as you can about how $\{k_{t+1}\}_{t=0}^{\infty}$ and the interest rate behave in these two economies.

[Figure 11.6: panel a: $c_t \approx .64$ until $t = 10$, then a jump down to $\approx .44$ and flat; panel b: c_t declines smoothly from $\approx .6$ toward $\approx .44$; panel c: $g_t = .2$ until $t = 10$, then $.4$.]

Solution:

Panel a: unanticipated. Before $t = 10$ nobody expects a change: the economy stays in the old steady state. At $t = 10$ the permanent, lump-sum-financed rise in g is a pure wealth effect (11.5(c)): consumption falls by exactly $\Delta g = .2$, and nothing else changes. $k_t = \bar{k}$ throughout and $\bar{R}_t = 1 + \rho$, $r_{t,t+1} = -\log \beta$ throughout.

Panel b: anticipated since 0. At $t = 0$ the present value of lump-sum taxes jumps, so consumption drops at once (wealth effect) and then keeps declining (section 11.9, figure 11.9.1). The household smooths consumption by building a stock of capital before $t = 10$ and running it down afterwards. So k rises from $\bar{k}$ to a peak at $t = 10$ and returns to $\bar{k}$ (the steady state does not depend on g). A declining consumption path needs $\bar{R}_{t+1} < 1 + \rho$, by (11.6.9): the interest rate $r_{t,t+1} = \log \bar{R}_{t+1}$ falls below $-\log \beta$ from $t = 0$, reaches its minimum between 9 and 10 when k peaks, and rises back afterwards.

Numerical exercise — see the companion code

Benchmark, $g = .2 \rightarrow .4$ at $t = 10$. Anticipated: $c_0 = .6092$, $c_{10} = .5390$, $c_{40} = .4446$; $k_{10} = 2.0985$ (maximum); minimum $\bar{R} = \bar{R}_{10} = 1.0008$; c strictly decreasing and $\bar{R} < 1 + \rho$ for 40 periods (checked). Surprise: the Newton path is $k_t = 1.4900$, $c = .6426$ before 10 and $.4426$ after, $\bar{R} = 1.0526$ throughout.

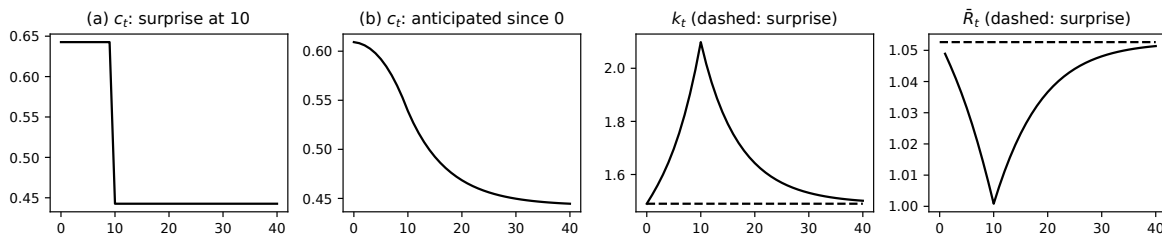


Figure 15: Exercise 11.17: panels (a) and (b) reproduced; capital and $\bar{R}$ under anticipation (solid) and surprise (dashed).

Habits, and the Invisible Hand (11.18–11.20)

Exercise 11.18 Habits and durability

Question

p. 468. A planner chooses sequences $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ to maximize $\sum_{t=0}^{\infty} \beta^t u(c_t - \alpha c_{t-1})$, $\alpha \in (-1, 1)$, $\beta \in (0, 1)$ subject to

$$k_{t+1} + c_t = f(k_t) + (1 - \delta)k_t, \quad \delta \in (0, 1), \quad c_t \geq 0$$

where $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$ if $\gamma > 0$ and $\gamma \neq 1$, $\log(x)$ if $\gamma = 1$, and (k_0, c_{-1}) are given initial conditions. Here $u' > 0$, $u'' < 0$, and $\lim_{c \downarrow 0} u'(c) = 0$. If $\alpha > 0$, it indicates that the consumer has a ‘habit’; if $\alpha < 0$, it indicates that the consumption good is somewhat durable.

- Find a complete set of first-order necessary conditions for the planner’s problem.
- Define an optimal *steady state*.
- Find optimal steady state values for $(k, f'(k), c)$.
- For given initial conditions (k_0, c_{-1}) , describe as completely as you can an algorithm for computing a path $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ that solves the planning problem.

Solution: *Misprint.* “ $\lim_{c \downarrow 0} u'(c) = 0$ ” should be $+\infty$ (Inada; for the stated u , $u'(x) = x^{-\gamma} \rightarrow \infty$). We assume $c_t - \alpha c_{t-1} > 0$ along the path.

(a) Lagrangian with multipliers $\beta^t \mu_t$:

$$\mathcal{L} = \sum_{t \geq 0} \beta^t \left\{ u(c_t - \alpha c_{t-1}) + \mu_t [f(k_t) + (1 - \delta)k_t - c_t - k_{t+1}] \right\}.$$

c_t appears in $u(c_t - \alpha c_{t-1})$ (date t) and in $u(c_{t+1} - \alpha c_t)$ (date $t + 1$). FOCs:

$$c_t : \quad \mu_t = u'(c_t - \alpha c_{t-1}) - \alpha \beta u'(c_{t+1} - \alpha c_t), \quad (11.18.1)$$

$$k_{t+1} : \quad \mu_t = \beta \mu_{t+1} [f'(k_{t+1}) + 1 - \delta], \quad (11.18.2)$$

$$k_{t+1} = f(k_t) + (1 - \delta)k_t - c_t, \quad (11.18.3)$$

for $t \geq 0$, with (k_0, c_{-1}) given and the transversality condition $\lim_t \beta^t \mu_t k_{t+1} = 0$. μ_t is the marginal value of goods at t : one more unit consumed raises utility today but, through the habit, lowers utility tomorrow ($\alpha > 0$), or raises it through durability ($\alpha < 0$).

(b) An optimal steady state is a triple (k, c, μ) that satisfies (11.18.1)–(11.18.3) with $k_t = k$, $c_t = c$, $\mu_t = \mu$ for all t . If $(k_0, c_{-1}) = (k, c)$, the optimal plan stays there.

(c) (11.18.2): $1 = \beta[f'(k) + 1 - \delta]$, so $f'(\bar{k}) = \rho + \delta$; (11.18.3): $\bar{c} = f(\bar{k}) - \delta\bar{k}$; (11.18.1): $\mu = (1 - \alpha\beta)u'((1 - \alpha)\bar{c}) > 0$ (because $|\alpha| < 1$). The steady state $(\bar{k}, f'(\bar{k}), \bar{c})$ is the same as without habits: habits and durability change the dynamics, not the long run.

(d) Substituting (11.18.1) into (11.18.2) gives one equation in consumption at four dates and capital:

$$u'(c_t - \alpha c_{t-1}) - \alpha\beta u'(c_{t+1} - \alpha c_t) = \beta[f'(k_{t+1}) + 1 - \delta] [u'(c_{t+1} - \alpha c_t) - \alpha\beta u'(c_{t+2} - \alpha c_{t+1})].$$

With (11.18.3), the system maps $s_t = (k_t, c_{t-1}, c_t, c_{t+1})$ into s_{t+1} : solve (11.18.3) for k_{t+1} , then the equation above for c_{t+2} . Two components are predetermined, (k_0, c_{-1}) , and two are free, (c_0, c_1) ; the terminal condition requires convergence to $(\bar{k}, \bar{c})$. Algorithms:

1. *Newton on the whole path* (used here). Pick S large. Unknowns $k_1, \dots, k_S$, with $k_{S+1} = k_{S+2} = \bar{k}$; $c_t = Y(k_t) - k_{t+1}$ for $t = 0, \dots, S+1$, c_{-1} given. Impose the Euler equation for $t = 0, \dots, S-1$: S equations in S unknowns. Each equation involves $k_{t-1}, \dots, k_{t+3}$, so the Jacobian is banded and cheap.
2. *Two-dimensional shooting*: guess (c_0, c_1) , iterate $s_{t+1} = F(s_t)$ forward, and adjust the guesses (by Newton on two unknowns; bisection no longer applies) until $(k_S, c_S) \approx (\bar{k}, \bar{c})$.
3. *Linear approximation*: linearise F at the steady state, $\hat{s}_{t+1} = J\hat{s}_t$. Generically J has two roots inside and two outside the unit circle, in pairs $(\lambda, 1/(\beta\lambda))$ as in section 11.10.1. Choose $(\hat{c}_0, \hat{c}_1)$ so that $\hat{s}_0$ lies in the span of the stable eigenvectors, given $(\hat{k}_0, \hat{c}_{-1})$; then $\hat{s}_t = V_s \Lambda_s^t w$ (solving stable roots backward and unstable roots forward).

Numerical exercise — see the companion code

$\gamma = 2$, benchmark $f, g = 0$; the book's habit parameter is written a in the code. Steady state $\bar{k} = 1.489956$, $f'(\bar{k}) = .252632$, $\bar{c} = .842645$ for every a ; $\mu = 2.9575$ ($a = .5$), 1.0709 ($a = -.3$). Eigenvalues of J : $a = .5$: $.5162, .8219, 1.2807, 2.0390$; $a = -.3$: $-3.5581, -.2958, .8309, 1.2668$. The negative pair makes consumption oscillate when the good is durable. Pairing $\lambda \cdot 1/(\beta\lambda)$ checked. Newton vs the stable-subspace solution for $k_0 = .99\bar{k}$, $c_{-1} = \bar{c}$: $\max |\Delta k| = 1.2 \times 10^{-5}$ ($a = .5$), 6.4×10^{-6} ($a = -.3$), against a deviation of .0149. At $a = 0$ the Newton path equals the standard model's. From $k_0 = .5\bar{k}$, $c_{-1} = f(k_0) - \delta k_0 = .7584$: $c_0 = .6475$ ($a = 0$), $.7018$ ($a = .5$: habit keeps consumption close to c_{-1} and slows accumulation, $k_{10} = 1.3428$ vs 1.3602), $.6158$ ($a = -.3$: consumption zig-zags, $.6158, .7012, .7120, \dots$).

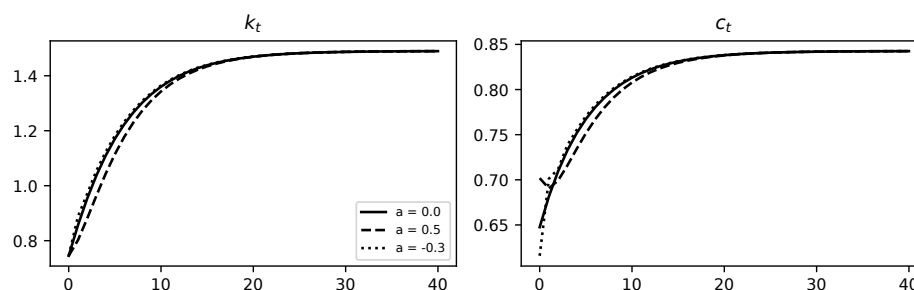


Figure 16: Exercise 11.18: paths from $k_0 = .5\bar{k}$ for $a = 0$ (solid), habit $a = .5$ (dashed), durability $a = -.3$ (dotted).

Exercise 11.19 The invisible hand, I

Question

p. 469. Please consider once again the “Foreseen jump in g ” experiment studied in section 11.9. This exercise asks you to put yourself in the shoes of the representative household and to think through the optimum problem that it faces within a competitive equilibrium at time 0.

- Please describe the signals that the market sends to the household *after* the new $\{g_t\}_{t=0}^{\infty}$ policy materializes at time 0. Please list all of the objects that “the market” (also known as “the invisible hand”) presents as *exogenous* to the representative household.
- Please describe how the “Big K ” part of a “Big K , little k ” argument is used to determine *all* of those objects exogenous to the household. **Hint:** This is accomplished by applying the shooting algorithm in section 11.9.
- Please explain thoroughly how the representative household chooses to respond to the signals presented to it by the market at time 0.
- Given the objects that the market presents to the representative household, please tell how you would use a shooting algorithm to compute the path of $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ chosen by the household.
- Please describe how to complete a “Big K , little k ” argument using your answers to parts **c** and **d**.

Solution: Experiment: benchmark economy at its steady state; at $t = 0$ it becomes known that g rises from .2 to .4 at $t = 10$, financed by lump-sum taxes.

(a) The household does not need to observe g_t . It sees: the time-0 price system $\{q_t, \eta_t, w_t\}_{t \geq 0}$ (prices of dated goods, rental rates, wages) and its tax bill $\{\tau_{ht}\}_{t \geq 0}$ (only $\sum_t q_t \tau_{ht}$ matters), with distorting taxes at zero. Exogenous to the household: $\{q_t, \eta_t, w_t, \tau_{ht}\}$ and its own k_0 . The news reaches it only through these objects: a higher present value of taxes and a new shape of q_t (lower interest rates).

(b) Big K : the economist (or the invisible hand) solves for the aggregate path $\{K_{t+1}\}$

by shooting on the aggregate Euler equation and feasibility,

$$u'(C_t) = \beta u'(C_{t+1})[f'(K_{t+1}) + 1 - \delta], \quad C_t = f(K_t) + (1 - \delta)K_t - g_t - K_{t+1},$$

from $K_0 = \bar{k}$ toward $\bar{k}$ with the new $\{g_t\}$. It then sets prices from (11.6.8): $q_t = \beta^t u'(C_t)/u'(C_0)$, $\eta_t = f'(K_t)$, $w_t = f(K_t) - K_t f'(K_t)$, and taxes with $\sum_t q_t \tau_{ht} = \sum_t q_t g_t$ (e.g. $\tau_{ht} = g_t$).

(c) Household problem at those prices: $\max \sum_t \beta^t u(c_t)$ s.t. $\sum_t q_t [c_t + k_{t+1} - (1 - \delta)k_t] \leq \sum_t q_t [\eta_t k_t + w_t - \tau_{ht}]$. Because the prices satisfy no arbitrage, $q_{t-1} = q_t(\eta_t + 1 - \delta)$, the coefficient of every k_t , $t \geq 1$, is zero. The household is indifferent about how it holds its wealth, and the budget collapses to

$$\sum_t q_t c_t \leq W_0 \equiv q_0(\eta_0 + 1 - \delta)k_0 + \sum_t q_t (w_t - \tau_{ht}).$$

FOC $\beta^t u'(c_t) = \mu q_t$ gives $c_t = (\mu q_t / \beta^t)^{-1/\gamma}$, and the budget pins down μ : $\mu^{-1/\gamma} \sum_t q_t (q_t / \beta^t)^{-1/\gamma} = W_0$. The response has two parts. *Wealth*: higher τ_h lowers W_0 and scales every c_t down. *Tilt*: q_t / β^t rises over $t \leq 10$ (interest rates below ρ), so c_t declines over time. Its capital holdings follow from the period budget once the timing of taxes is fixed.

(d) Shooting at *given* prices: guess c_0 ; iterate $u'(c_{t+1}) = u'(c_t) q_{t+1} / (\beta q_t)$ and $k_{t+1} = (\eta_t + 1 - \delta)k_t + w_t - \tau_{ht} - c_t$ (with $\tau_{ht} = g_t$); raise c_0 if k_S ends above the target and lower it if below, until $k_S \approx \bar{k}$ (equivalently, until the present-value budget holds with equality). Unlike the algorithm of (b), the prices do not respond to the household's k .

(e) The prices of (b) are equilibrium prices if the household's optimal choices at those prices, from (c)–(d), reproduce the aggregates: $c_t = C_t$ and $k_t = K_t$ for all t . Then markets clear and every agent optimises. This holds here because the aggregate Euler equation of (b) *is* the household's Euler equation evaluated at $c = C$, $k = K$: the shooting in (b) finds the fixed point of the map from a conjectured aggregate law of motion to the household's choices.

Numerical exercise — see the companion code

Big K by Newton; little k in closed form at those prices: $\mu = 2.694141$ (with $q_0 = 1$); household $c_0 = .609242 = C_0$; $c_t = C_t$ to 10^{-6} relative for all t ; $k_t = K_t$ over 60 periods to 2.8×10^{-10} (with $\tau_{ht} = g_t$); no-arbitrage $q_t/q_{t+1} = \eta_{t+1} + 1 - \delta$ checked.

Exercise 11.20 The invisible hand, II

Question

p. 470. Please consider again the “Foreseen jump in τ_n ” experiment in section 11.12. Like the previous problem, this one puts you into the shoes of the representative household and asks you to think through the optimum problem that it faces within a competitive equilibrium with distorting taxes at time 0.

a. Please describe the signals that the market sends to the household *after* the new $\{\tau_{nt}\}_{t=0}^{\infty}$ policy materializes at time 0. Please list all objects that “the market” presents as *exogenous* to the representative household.

b. Keeping in mind that there is a “Big K , little k ” argument in the background, please provide a complete explanation for why the household chooses the paths of $\{c_t, n_t, k_t\}_{t=0}^{\infty}$ displayed in figure 11.12.3.

Solution: Section 11.12: $U = \ln c + B(1 - n)$, $B = 3$, $F = k^\alpha n^{1-\alpha}$, $g = .2$, $\tau_c = \tau_k = 0$; at $t = 0$ it becomes known that τ_n rises from 0 to .2 at $t = 10$; lump-sum transfers balance the budget.

(a) The household sees the price system $\{q_t, \eta_t, w_t\}_{t \geq 0}$, the announced labour-tax path $\{\tau_{nt}\}$ (0 for $t \leq 9$, .2 after), $\tau_{ct} = \tau_{kt} = 0$, and its lump-sum taxes or transfers $\{\tau_{ht}\}$ (through their present value). These, with k_0 , are exogenous to it.

(b) Household problem: $\max \sum_t \beta^t [\ln c_t + B(1 - n_t)]$ s.t. $\sum_t q_t [c_t + k_{t+1} - (1 - \delta)k_t] \leq \sum_t q_t [\eta_t k_t + (1 - \tau_{nt})w_t n_t - \tau_{ht}]$. FOCs:

$$\frac{\beta^t}{c_t} = \mu q_t, \quad \beta^t B = \mu q_t (1 - \tau_{nt}) w_t \quad (0 < n_t < 1), \quad \frac{q_t}{q_{t+1}} = 1 + \eta_{t+1} - \delta.$$

Dividing: $Bc_t = (1 - \tau_{nt})w_t$ (11.12.15c), and $c_{t+1} = \beta \bar{R}_{t+1} c_t$ (11.12.15a). *Key feature:* utility is linear in leisure. At prices that satisfy these conditions, the marginal value of an hour of work is the same at every date, so the household is indifferent about *when* it works; only the present value of its labour income matters. Its consumption is pinned down by the after-tax wage, and its hours and capital are pinned down by market clearing, i.e. by the Big K path. The figure then follows:

- *Long run.* (11.12.12) fixes $k/n = \tilde{k}$ independently of τ_n , so the pre-tax wage returns to its old value, and $c = (1 - \tau_n)w/B$ falls by 20%. By (11.12.14) n falls, and $k = \tilde{k}n$ falls with it.
- *Before $t = 10$.* Work is cheap in utility terms relative to after 10: the after-tax wage is high now and will be low later. The household works more before 10 (n rises) and invests the extra output (k rises): intertemporal substitution of labour. Consumption stays essentially at its old level. By the Euler equation and $Bc_t = w_t$, a flat c requires a constant $\bar{R}$ and wage, i.e. a constant k/n , so k and n rise together. (Exactly, c_t is a hair below its old level and rises toward it: $c_{t+1} = \beta \bar{R}(\kappa(c_{t+1}))c_t$ with κ increasing in c is a stable map toward $\bar{c}$.)
- *At $t = 10$.* Hours drop sharply. The capital built up raises k/n and the pre-tax wage, so the after-tax wage falls by less than 20% at first. $\bar{R}_{10}$ falls below 1 (low

f' at a high k/n), so consumption falls.

- After $t = 10$. Capital is run down, k/n falls back to $\tilde{k}$, the wage declines to its old value, $\bar{R}$ rises back to $1 + \rho$, c declines to its new level and n recovers to its new, lower steady state.

Big K , little k : the particular paths of n_t and k_t are those that, aggregated, generate the prices at which the household is willing to supply them.

Numerical exercise — see the companion code

Old steady state $k = .8041$, $c = .2547$, $n = .5397$; new $k = .7140$, $c = .2038$, $n = .4792$. Before 10, $c_t/c_{\text{old}} - 1 \in [-1.15 \times 10^{-3}, -3.2 \times 10^{-5}]$; $n_9 = .6136$, $k_{10} = .9764$, $n_{10} = .4087$, $w_{10} = .8931$, $\bar{R}_{10} = .9841$ (figure 11.12.3 reproduced). Newton on (11.12.15) vs shooting on c_0 (solving for k_{t+1}/n_{t+1} each period): paths agree to 10^{-7} ; $Bc_t = (1 - \tau_{nt})w_t$ holds on the path.

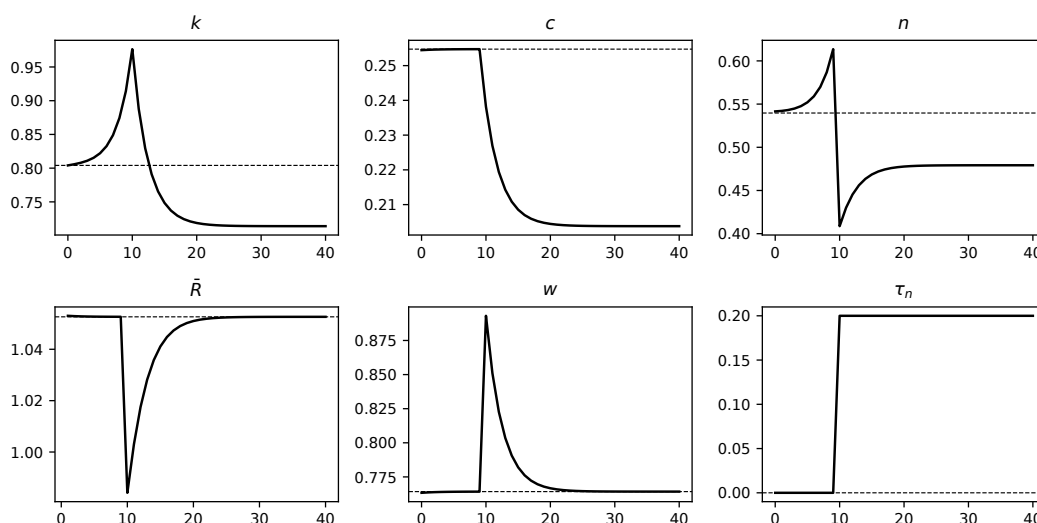


Figure 17: Exercise 11.20: response to the foreseen rise in τ_n at $t = 10$ (dashed: old steady state).

Companion code

Every numerical result above is reproduced by the notebook `ls_ch11.ipynb` in this folder (executed, outputs saved). Each claim is checked by an independent route with an `assert`.

Section	What it checks
Shared	Policy paths $(g, \tau_c, \tau_k, \psi, \text{levy})$; damped Newton with banded finite-difference Jacobian on (11.6.3); shooting of section 11.6.3 by bisection on c_0 ; linearisation (11.10.8) from numerical derivatives of H ; prices (11.6.8). The ch. 5/7 LQ toolkit is not needed.
11.1–11.4	Newton vs shooting; PV budgets by <code>brentq</code> ; Laffer curves (steady-state τ_k ; levy at $T = 10$); sequential government budget (11.4.2).
11.5–11.9	Flat paths after a surprise g ; open-economy budgets; welfare rankings; linear vs Newton (error first order); λ_2 root vs corrected (11.10.16); <code>linprog</code> corners.
11.10–11.17	$r_{t,t+1} = \log \bar{R}_{t+1}$; shapes (monotonicity, concavity) of c , k , $\bar{R}$, $r_{0,t}$; VFI vs closed form; revenue-neutral $\bar{\tau}_k$.
11.18–11.20	Habit Newton vs stable subspace, root pairing; household problem at equilibrium prices vs aggregates; elastic-labour Newton vs shooting.

Figures are saved to `fig/ls11_*.pdf` with TrueType (not Type 3) fonts.